Network Theory

Detailed Solutions

GATE INSTRUMENTATION ENGINEERING

PREPFUSION

Previous Year Questions Solutions

Contents

SOLUTIONS

I

Electrostatics

Topics

1. Coulomb's Law and Superposition Principle
2. Electric Field, Electric Field Intensity calculation in various Configurations: Due to Point, Line, Plane and Spherical Charge Distributions
3. Equipotential Surfaces and Electric Dipoles
4. Electric Flux Density and Gauss's Law
5. Divergence and Electric Potential
6. Electric Field Effect of Dielectric Medium- a)Boundary conditions b)method of image charges
7. Capacitance of Simple Configurations

Q1.26.1 NAT 2026 2M Ans: 0.40

Given,

$$\rho_v = \frac{3}{\pi} \times 10^{-6}\,\text{C/m}^3, \qquad R = 2\,\text{m}, \qquad q = \pi \times 8.854 \times 10^{-12}\,\text{C}$$

$$\text{Initial position: } (-3, 0, -4), \qquad \text{Final position: } (0, 0, 4)$$

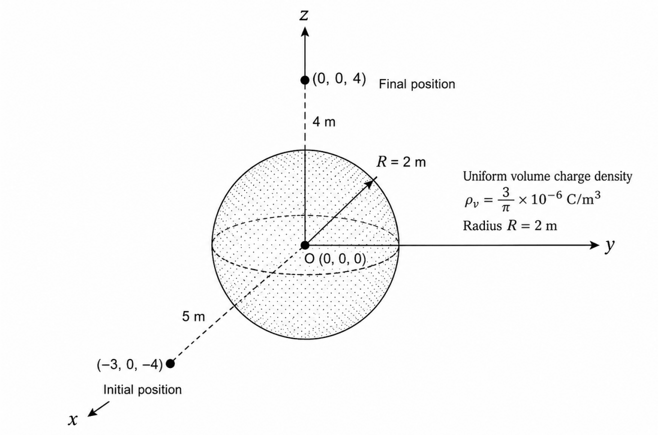

Fig. Solution Q1.26.1

Since $\varepsilon_r = 1$,

$$\varepsilon = \varepsilon_0$$

Step 1: Distances from the centre

Initial distance:

$$r_1 = \sqrt{(-3)^2 + 0^2 + (-4)^2} = \sqrt{25} = 5\,\text{m}$$

Final distance:

$$r_2 = \sqrt{0^2 + 0^2 + 4^2} = 4\,\text{m}$$

Since $r_1, r_2 > R$, both points lie outside the sphere.

Step 2: Total charge enclosed

$$Q = \rho_v \left(\frac{4}{3}\pi R^3\right)$$

$$Q = \left(\frac{3}{\pi} \times 10^{-6}\right)\left(\frac{4}{3}\pi \times 2^3\right) = 32 \times 10^{-6}\,\text{C}$$

$$Q = 3.2 \times 10^{-5}\,\text{C}$$

Step 3: Potential difference

For $r > R$, the sphere behaves like a point charge at the origin: $V(r) = \frac{Q}{4\pi\varepsilon_0 r}$

Hence,

$$V_2 - V_1 = \frac{Q}{4\pi\varepsilon_0}\left(\frac{1}{r_2} - \frac{1}{r_1}\right)$$

$$V_2 - V_1 = \frac{Q}{4\pi\varepsilon_0}\left(\frac{1}{4} - \frac{1}{5}\right) = \frac{Q}{80\pi\varepsilon_0}$$

Step 4: Work done

$$W = q(V_2 - V_1)$$

$$W = \left(\pi \times 8.854 \times 10^{-12}\right)\left(\frac{3.2 \times 10^{-5}}{80\pi \times 8.854 \times 10^{-12}}\right)$$

$$W = \frac{3.2 \times 10^{-5}}{80} = 4 \times 10^{-7}\,\text{J}$$

$$\boxed{0.40}$$

Key Points

- Electric field:

$$E(r) = \begin{cases} \dfrac{\rho_v r}{3\varepsilon}, & r < R \\ \dfrac{Q}{4\pi\varepsilon r^2}, & r \geq R \end{cases}$$

- Electric potential:

$$V(r) = \begin{cases} \dfrac{Q}{8\pi\varepsilon R}\left(3 - \dfrac{r^2}{R^2}\right), & r < R \\ \dfrac{Q}{4\pi\varepsilon r}, & r \geq R \end{cases}$$

Mistakes to be avoided

- Always check whether the initial and final points lie inside or outside the charged sphere before applying

$$V(r) = \frac{Q}{4\pi\varepsilon r}.$$

This expression is valid only for $r \geq R$.

Q1.26.2 MCQ 2026 1M Ans: C

A uniformly charged ring is symmetric about its centre. For every charge element dq on the ring, there is a diametrically opposite element producing an equal and opposite electric field at the centre. Hence the net electric field at the centre is

$$\vec{E}_{\text{center}} = 0$$

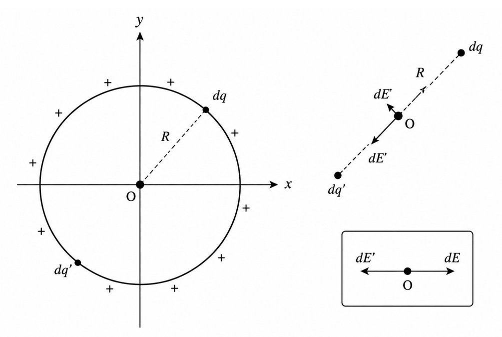

Fig. Solution Q1.26.2

The force on a point charge q placed at the centre is

$$\vec{F} = q\vec{E} = 0$$

Hence, the magnitude of the force is

$$\boxed{C}$$

Q1.26.3 MCQ 2026 1M Ans: B ● ● ○

Given,

$$\vec{v} = 5\hat{x}$$

$$\vec{E} = 4\hat{y}$$

$$\vec{B} = -6\hat{z}$$

and the charge q is positive.
The Lorentz force acting on the charge is

$$\vec{F} = q\left(\vec{E} + \vec{v} \times \vec{B}\right)$$

First, evaluate the magnetic force component.

$$\vec{v} \times \vec{B} = (5\hat{x}) \times (-6\hat{z})$$

Using

$$\hat{x} \times \hat{z} = -\hat{y}$$

we get

$$\vec{v} \times \vec{B} = (5)(-6)(-\hat{y}) = 30\hat{y}$$

Substituting into the Lorentz force equation,

$$\vec{F} = q\,(4\hat{y} + 30\hat{y})$$

$$\vec{F} = 34q\,\hat{y}$$

Hence, the force acts along the positive y-direction, i.e., along the direction of the electric field and perpendicular to the magnetic field.

$$\boxed{B}$$

Key Points

- Lorentz force:
$$\vec{F} = q\left(\vec{E} + \vec{v} \times \vec{B}\right)$$
- Unit vector identities:
$$\hat{x} \times \hat{y} = \hat{z}, \qquad \hat{y} \times \hat{z} = \hat{x}, \qquad \hat{z} \times \hat{x} = \hat{y}$$
- Reversing the order changes the sign of the cross product.
- For a positive charge, the force direction is the same as $\vec{E} + \vec{v} \times \vec{B}$.

Mistakes to be avoided

- Do not use $\hat{x} \times \hat{z} = \hat{y}$; the correct relation is $\hat{x} \times \hat{z} = -\hat{y}$.
- Do not forget to include both the electric and magnetic force components.
- Remember that the direction reverses for a negative charge.

Q1.25.1 MCQ 2025 2M Ans: C ● ● ○

The original air-filled coaxial capacitor has capacitance

$$C_0 = \frac{2\pi\varepsilon_0 L}{\ln(b/a)}$$

When dielectric fills 50% of the annular region angularly, the capacitor is equivalent to two capacitors in parallel: air-filled half and dielectric-filled half

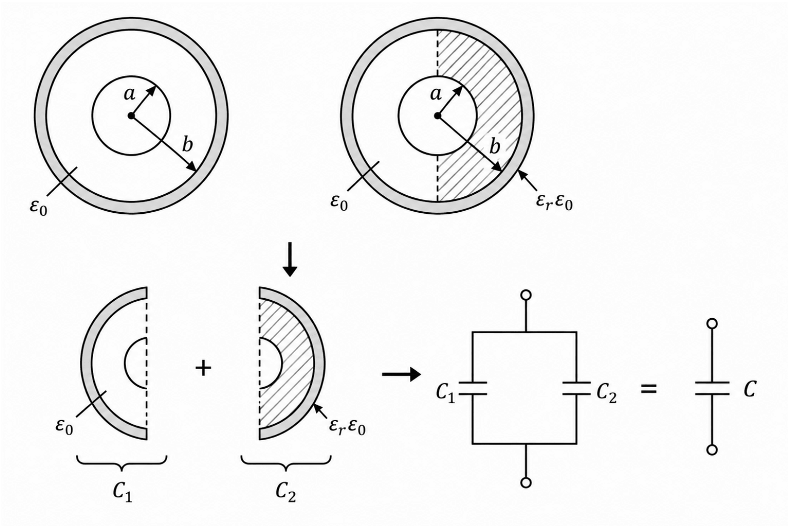

Fig. Solution Q1.25.1

Hence,

$$C_1 = \frac{1}{2}C_0$$

and

$$C_2 = \frac{1}{2}\left(\frac{2\pi\varepsilon_r\varepsilon_0 L}{\ln(b/a)}\right) = \frac{\varepsilon_r}{2}C_0$$

Therefore, total capacitance is

$$C = C_1 + C_2 = \frac{1+\varepsilon_r}{2}C_0$$

Given that

$$C = 5C_0$$

So,

$$\frac{1+\varepsilon_r}{2} = 5 \Rightarrow 1+\varepsilon_r = 10 \Rightarrow \varepsilon_r = 9$$

$$\boxed{C}$$

Key Points

- For annular filling fraction f,

$$C = \left[(1-f) + f\varepsilon_r\right]C_0$$

- Dielectric inserted over part of the annular region:

$$\boxed{\text{Capacitors are in parallel}}$$

- Dielectric inserted over part of the radial thickness:

$$\boxed{\text{Capacitors are in series}}$$

Q2.25.2 MCQ 2025 2M Ans: B,D ● ● ○

Given,

$$\vec{F} = \frac{\hat{a}_R}{R^n}$$

For a purely radial vector field in spherical coordinates,

$$\nabla \cdot \vec{F} = \frac{1}{R^2}\frac{d}{dR}\left(R^2 F_R\right)$$

Here,

$$F_R = \frac{1}{R^n}$$

So,

$$\nabla \cdot \vec{F} = \frac{1}{R^2}\frac{d}{dR}\left(R^{2-n}\right)$$

$$= \frac{1}{R^2}(2-n)R^{1-n} = (2-n)R^{-n-1}$$

Now check the given values of n.
For $n = -2$:

$$\nabla \cdot \vec{F} = (2+2)R = 4R$$

depends on R.

For $n = -1$:

$$\nabla \cdot \vec{F} = (2+1)R^0 = 3$$

independent of R.

For $n = 1$:

$$\nabla \cdot \vec{F} = (2-1)R^{-2} = \frac{1}{R^2}$$

depends on R, so it is not independent of R.

For $n = 2$:

$$\nabla \cdot \vec{F} = (2-2)R^{-3} = 0$$

which is also independent of R. Hence, the correct options are

$$\boxed{B, D}$$

Key Points

- Divergence of a pure radial vector field in Cartesian coordinates:

$$\nabla \cdot \vec{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

- Divergence of a pure radial vector field in cylindrical coordinates:

$$\nabla \cdot \vec{A} = \frac{1}{\rho}\frac{d}{d\rho}\left(\rho A_\rho\right)$$

- Divergence of a pure radial vector field in spherical coordinates:

$$\nabla \cdot \vec{A} = \frac{1}{R^2}\frac{d}{dR}\left(R^2 A_R\right)$$

Q1.25.3 MCQ | 2025 | 1M | Ans: C | ● ○ ○

Given,

$$\rho = \begin{cases} -3\rho_0, & 0 \le R \le a \\ 0, & R > a \end{cases} \qquad (\rho_0 > 0)$$

For $0 \le R \le a$:
Using Gauss's law,

$$\oint \vec{E} \cdot d\vec{S} = \frac{Q_{\text{enc}}}{\varepsilon_0}$$

For a Gaussian sphere of radius R,

$$Q_{\text{enc}} = (-3\rho_0)\left(\frac{4}{3}\pi R^3\right) = -4\pi\rho_0 R^3$$

So,

$$E(4\pi R^2) = \frac{-4\pi\rho_0 R^3}{\varepsilon_0} \Rightarrow E_R = -\frac{\rho_0 R}{\varepsilon_0}$$

Thus,

$$E_R = -\frac{\rho_0 R}{\varepsilon_0}, \qquad 0 \le R \le a$$

At $R = a$,

$$E_R(a) = -\frac{\rho_0 a}{\varepsilon_0}$$

For $R > a$:
Total charge enclosed is

$$Q = (-3\rho_0)\left(\frac{4}{3}\pi a^3\right) = -4\pi\rho_0 a^3$$

Again by Gauss's law,

$$E(4\pi R^2) = \frac{-4\pi\rho_0 a^3}{\varepsilon_0} \Rightarrow E_R = -\frac{\rho_0 a^3}{\varepsilon_0 R^2}$$

Thus,

$$E_R = -\frac{\rho_0 a^3}{\varepsilon_0 R^2}, \qquad R > a$$

At $R = a$,

$$E_R(a) = -\frac{\rho_0 a^3}{\varepsilon_0 a^2} = -\frac{\rho_0 a}{\varepsilon_0}$$

So the electric field is continuous at $R = a$.
Hence, the graph is:
i) linear for $R < a$, ii) varies as $-1/R^2$ for $R > a$, iii) and tends to zero as $R \to \infty$.

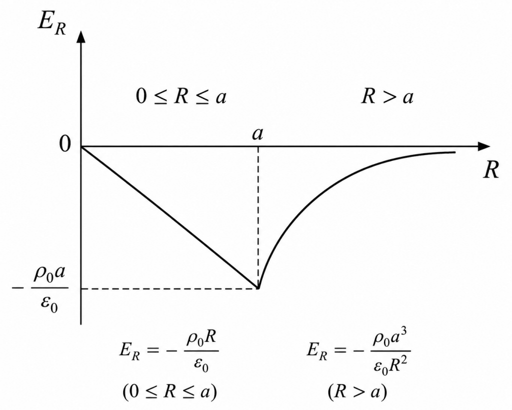

Fig. Solution Q1.25.3

Therefore, the correct plot is Fig. (iii).

$$\boxed{C}$$

Q1.24.1 NAT 2024 2M Ans: -1.6

Given,

$$Q \text{ at } (-1,0,0), \qquad Q \text{ at } (1,0,0), \qquad \alpha Q \text{ at } (0,-1,0)$$

The electric field is required to be zero at P(0,0.5,0)

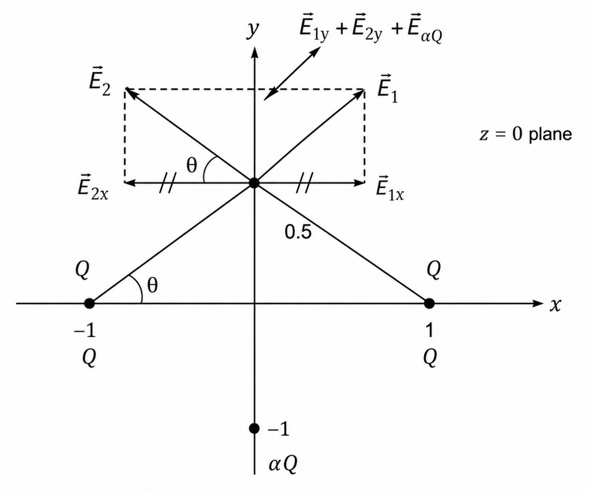

Fig. Solution Q1.24.1

Electric field due to the charges Q

For the charge at $(-1,0,0)$,

$$\vec{r}_1 = \hat{\imath} + \frac{1}{2}\hat{\jmath}, \qquad |\vec{r}_1| = \frac{\sqrt{5}}{2}$$

Hence,

$$\vec{E}_1 = kQ\frac{\vec{r}_1}{|\vec{r}_1|^3} = \frac{8kQ}{5\sqrt{5}}\hat{\imath} + \frac{4kQ}{5\sqrt{5}}\hat{\jmath}$$

Similarly, for the charge at $(1,0,0)$,

$$\vec{r}_2 = -\hat{\imath} + \frac{1}{2}\hat{\jmath}, \qquad |\vec{r}_2| = \frac{\sqrt{5}}{2}$$

$$\vec{E}_2 = -\frac{8kQ}{5\sqrt{5}}\hat{\imath} + \frac{4kQ}{5\sqrt{5}}\hat{\jmath}$$

Therefore,

$$\vec{E}_1 + \vec{E}_2 = \frac{8kQ}{5\sqrt{5}}\hat{\jmath}$$

The x-components cancel due to symmetry.

Electric field due to αQ

For the charge at $(0,-1,0)$,

$$\vec{r}_3 = 1.5\,\hat{\jmath} = \frac{3}{2}\hat{\jmath}, \qquad |\vec{r}_3| = \frac{3}{2}$$

Hence,

$$\vec{E}_3 = k\frac{\alpha Q}{(3/2)^2}\hat{j} = \frac{4k\alpha Q}{9}\hat{j}$$

Condition for zero electric field

$$\vec{E}_1 + \vec{E}_2 + \vec{E}_3 = 0$$

$$\frac{8}{5\sqrt{5}} + \frac{4\alpha}{9} = 0$$

$$\alpha = -\frac{18}{5\sqrt{5}} = -1.6099$$

$$\boxed{-1.6}$$

Mistakes to be avoided

- For NAT questions, always report the final answer with the precision specified in the question. Here, the answer must be rounded to one decimal place:

Q1.22.1 MCQ 2022 2M Ans: C ● ● ●

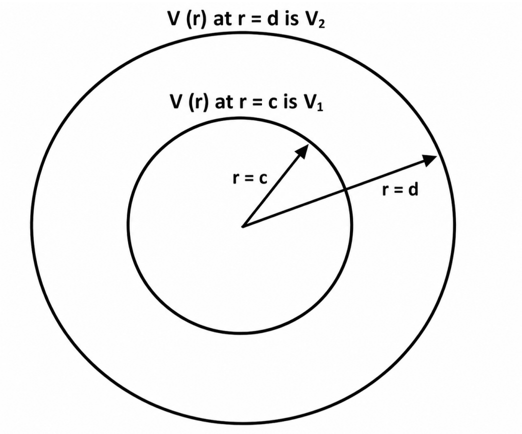

Fig. Solution Q1.22.1

For the region between the two conducting spherical shells ($c < r < d$), there is no charge present. Therefore,

$$\nabla^2 V = 0$$

For spherical symmetry,

$$\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{dV}{dr}\right) = 0$$

Integrating once,

$$r^2\frac{dV}{dr} = A$$

$$\Rightarrow \frac{dV}{dr} = \frac{A}{r^2}$$

Integrating again,

$$V(r) = -\frac{A}{r} + B$$

or,

$$V(r) = \frac{C}{r} + D$$

where C and D are constants.
Applying the boundary conditions,

$$V(c) = V_1, \qquad V(d) = V_2$$

So,

$$V_1 = \frac{C}{c} + D, \qquad V_2 = \frac{C}{d} + D$$

Subtracting,

$$V_1 - V_2 = C\left(\frac{1}{c} - \frac{1}{d}\right) = C\frac{d-c}{cd}$$

$$\Rightarrow C = \frac{cd(V_1 - V_2)}{d-c}$$

Substituting into $V_1 = \frac{C}{c} + D$,

$$D = V_1 - \frac{d(V_1 - V_2)}{d-c} = \frac{dV_2 - cV_1}{d-c}$$

Hence,

$$V(r) = \frac{cd(V_1 - V_2)}{(d-c)r} + \frac{dV_2 - cV_1}{d-c}$$

or equivalently,

$$V(r) = \frac{cd(V_1 - V_2)}{(d-c)r} - \frac{V_1 c - V_2 d}{d-c}$$

Hence,

$$\boxed{C}$$

Key Points

- Poisson's equation:

$$\nabla^2 V = -\frac{\rho_v}{\varepsilon}$$

  In a charge-free region ($\rho_v = 0$), it reduces to Laplace's equation:

$$\boxed{\nabla^2 V = 0}$$

- The Laplacian in spherical coordinates is

$$\nabla^2 V = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial V}{\partial r}\right) + \frac{1}{r^2 \sin\theta}\frac{\partial}{\partial \theta}\left(\sin\theta \frac{\partial V}{\partial \theta}\right) + \frac{1}{r^2 \sin^2\theta}\frac{\partial^2 V}{\partial \phi^2}$$

  For spherical symmetry, $V = V(r)$, so the θ and ϕ terms vanish:

$$\boxed{\frac{1}{r^2}\frac{d}{dr}\left(r^2 \frac{dV}{dr}\right) = 0}$$

Q1.21.1 NAT 2021 1M Ans: 9.00

Given,

$$Q_1 = 1\,\mu\text{C} = 10^{-6}\,\text{C}, \qquad Q_2 = 10\,\mu\text{C} = 10^{-5}\,\text{C}$$

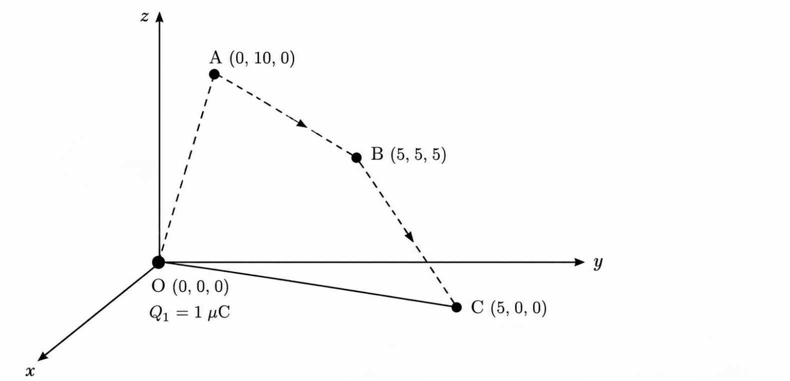

Fig. Solution Q1.21.1

The charge Q_2 is moved from $(0, 10, 0)$ to $(5, 5, 5)$ and finally to $(5, 0, 0)$. Since the electrostatic field is conservative, the work done depends only on the initial and final positions.
Initial distance from the charge at the origin:

$$r_i = \sqrt{0^2 + 10^2 + 0^2} = 10\,\text{m}$$

Final distance from the charge at the origin:

$$r_f = \sqrt{5^2 + 0^2 + 0^2} = 5\,\text{m}$$

Potential due to Q_1 at distance r is

$$V(r) = \frac{1}{4\pi\varepsilon_0}\frac{Q_1}{r}$$

Hence,

$$V_i = \frac{9 \times 10^9 \times 10^{-6}}{10} = 900\,\text{V}$$

and

$$V_f = \frac{9 \times 10^9 \times 10^{-6}}{5} = 1800\,\text{V}$$

Therefore, work done is

$$W = Q_2(V_f - V_i)$$

$$W = 10^{-5}(1800 - 900) = 9 \times 10^{-3}\,\text{J}$$

$$\boxed{9.00}$$

Q1.21.2 NAT 2021 2M Ans: 5

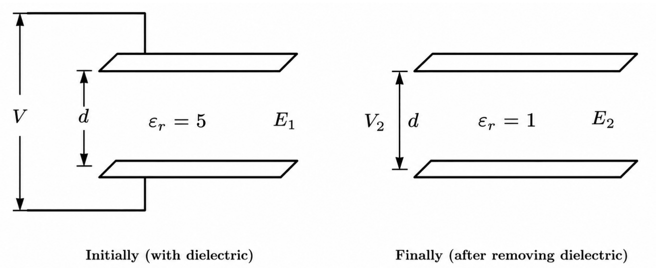

Fig. Solution Q1.21.2

The capacitor is initially charged to a potential difference V and then disconnected from the source. Hence,

$$Q = \text{constant}$$

Initially, the dielectric completely fills the gap, so

$$C_1 = \varepsilon_r C_0 = 5C_0$$

and

$$E_1 = \frac{V}{d}$$

After removing the dielectric, the capacitance becomes

$$C_2 = C_0$$

Since charge remains constant,

$$Q = C_1 V = 5C_0 V$$

The new voltage is

$$V_2 = \frac{Q}{C_2} = \frac{5C_0 V}{C_0} = 5V$$

Hence the new electric field is

$$E_2 = \frac{V_2}{d} = \frac{5V}{d}$$

Therefore,

$$\frac{E_2}{E_1} = \frac{5V/d}{V/d} = 5$$

$$\boxed{5}$$

Key Points

- Effect of changing/removing the dielectric:

Quantity	Source Disconnected	Source Connected
Q	Constant	Changes
V	Changes	Constant
E	$\propto \frac{1}{\varepsilon_r}$	Constant
C	$\propto \varepsilon_r$	$\propto \varepsilon_r$
U	$\propto \frac{1}{\varepsilon_r}$	$\propto \varepsilon_r$

Mistakes to be avoided

- Read the question carefully and check whether the source is connected or disconnected after changing/removing the dielectric.

Q1.20.1 MCQ | 2020 | 2M | Ans: D | ● ● ○

Given,

$$\vec{D} = 15\hat{a}_r + 2r\hat{a}_\phi - 3rz\hat{a}_z \quad \text{C/m}^2$$

The electric flux through the closed cylindrical surface is obtained using the divergence theorem:

$$\oint_S \vec{D} \cdot d\vec{S} = \iiint_V (\nabla \cdot \vec{D})\, dV$$

For cylindrical coordinates,

$$\nabla \cdot \vec{D} = \frac{1}{r}\frac{\partial}{\partial r}(rD_r) + \frac{1}{r}\frac{\partial D_\phi}{\partial \phi} + \frac{\partial D_z}{\partial z}$$

Here,

$$D_r = 15, \qquad D_\phi = 2r, \qquad D_z = -3rz$$

Hence,

$$\nabla \cdot \vec{D} = \frac{1}{r}\frac{\partial}{\partial r}(15r) + \frac{1}{r}\frac{\partial (2r)}{\partial \phi} + \frac{\partial(-3rz)}{\partial z} = \frac{15}{r} - 3r$$

For the solid cylinder,

$$0 \le r \le 3, \qquad 0 \le \phi \le 2\pi, \qquad 0 \le z \le 5$$

and

$$dV = r\,dr\,d\phi\,dz$$

Therefore,

$$\Phi = \iiint_V \left(\frac{15}{r} - 3r\right) r\,dr\,d\phi\,dz = \int_0^5 \int_0^{2\pi} \int_0^3 (15 - 3r^2)\,dr\,d\phi\,dz$$

Now,

$$\int_0^3 (15 - 3r^2)\,dr = \left[15r - r^3\right]_0^3 = 45 - 27 = 18$$

Thus,

$$\Phi = 18 \int_0^{2\pi} d\phi \int_0^5 dz = 18(2\pi)(5) = 180\pi$$

Hence,

$$\boxed{D}$$

Mistakes to be avoided

- In complex calculations, be careful in each step.

Q1.20.2 MCQ 2020 2M Ans: D ● ● ○

Given,

$$\vec{E} = 2r\,\hat{a}_r + \frac{3}{r}\,\hat{a}_\phi + 6\,\hat{a}_z$$

The volume charge density is

$$\rho_v = \nabla \cdot \vec{D} \qquad where \qquad \vec{D} = \varepsilon\vec{E} = \varepsilon_r\varepsilon_0\vec{E}$$

Hence,

$$\rho_v = \varepsilon_r\varepsilon_0(\nabla \cdot \vec{E})$$

For cylindrical coordinates,

$$\nabla \cdot \vec{E} = \frac{1}{r}\frac{\partial (rE_r)}{\partial r} + \frac{1}{r}\frac{\partial E_\phi}{\partial \phi} + \frac{\partial E_z}{\partial z}$$

Substituting,

$$E_r = 2r, \qquad E_\phi = \frac{3}{r}, \qquad E_z = 6$$

gives

$$\nabla \cdot \vec{E} = \frac{1}{r}\frac{\partial (2r^2)}{\partial r} + 0 + 0 = \frac{1}{r}(4r) = 4$$

Therefore,

$$\rho_v = \varepsilon_r\varepsilon_0(4) = 2.25 \times 4\,\varepsilon_0 = 9\varepsilon_0$$

$$\boxed{D}$$

Q1.20.3 MCQ 2020 2M Ans: C

For a conservative field,

$$\nabla \times \vec{F} = 0$$

Given,

$$\vec{F} = \hat{a}_x(5y - k_1 z) + \hat{a}_y(3z + k_2 x) + \hat{a}_z(k_3 y - 4x)$$

Let,

$$F_x = 5y - k_1 z, \qquad F_y = 3z + k_2 x, \qquad F_z = k_3 y - 4x$$

Now,

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{a}_x & \hat{a}_y & \hat{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_x & F_y & F_z \end{vmatrix}$$

For $\hat{a}_x$ component,

$$\frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z} = k_3 - 3 = 0$$

$$k_3 = 3$$

For $\hat{a}_y$ component,

$$\frac{\partial F_x}{\partial z} - \frac{\partial F_z}{\partial x} = (-k_1) - (-4) = 0$$

$$k_1 = 4$$

For $\hat{a}_z$ component,

$$\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} = k_2 - 5 = 0$$

$$k_2 = 5$$

Therefore,

$$k_1 = 4, \qquad k_2 = 5, \qquad k_3 = 3$$

$$\boxed{C}$$

Key Points

- Conservative field condition:
$$\nabla \times \vec{F} = 0$$
- Each component of curl must be individually equal to zero.

Q1.18.1 NAT 2018 2M Ans: 2.53

Given,

$$C_0 = 60\,\text{pF}$$

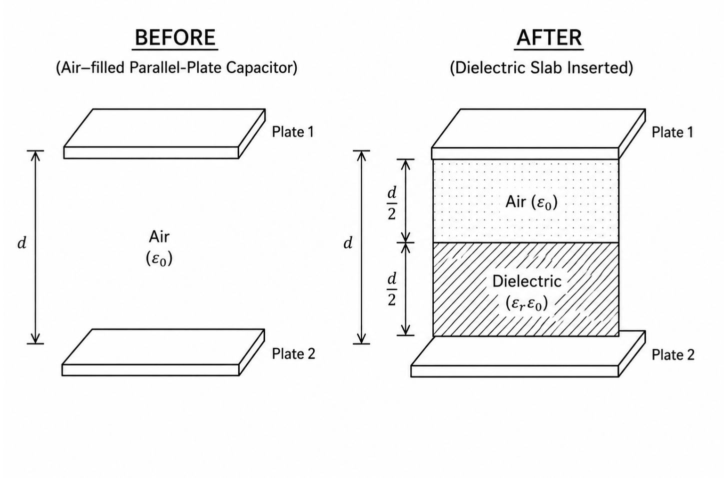

Fig. Solution Q1.18.1

A dielectric slab of thickness

$$t = \frac{d}{2}$$

is inserted between the plates covering the entire area.
The new capacitance is

$$C = 86\,\text{pF}$$

Let the relative permittivity of the dielectric be ε_r.
The arrangement is equivalent to two capacitors in series:
i) air layer of thickness $\frac{d}{2}$
ii) dielectric layer of thickness $\frac{d}{2}$
For the air-filled part,

$$C_1 = \frac{\varepsilon_0 A}{d/2} = \frac{2\varepsilon_0 A}{d} = 2C_0 = 120\,\text{pF}$$

For the dielectric-filled part,

$$C_2 = \frac{\varepsilon_r\varepsilon_0 A}{d/2} = \frac{2\varepsilon_r\varepsilon_0 A}{d} = 2\varepsilon_r C_0 = 120\varepsilon_r\,\text{pF}$$

Since C_1 and C_2 are in series,

$$C_{\text{eq}} = \frac{C_1 C_2}{C_1 + C_2}$$

Substituting,

$$86 = \frac{120 \times 120\varepsilon_r}{120 + 120\varepsilon_r}$$

$$86 = \frac{120\varepsilon_r}{1 + \varepsilon_r}$$

$$\varepsilon_r = \frac{86}{34} = 2.5294$$

Hence,

$$\boxed{2.53}$$

Q1.18.2 MCQ 2018 1M Ans: D

Given,

$$q = 1\,\text{nC} = 10^{-9}\,\text{C}, \qquad \text{located at } (0,0,0.2)$$

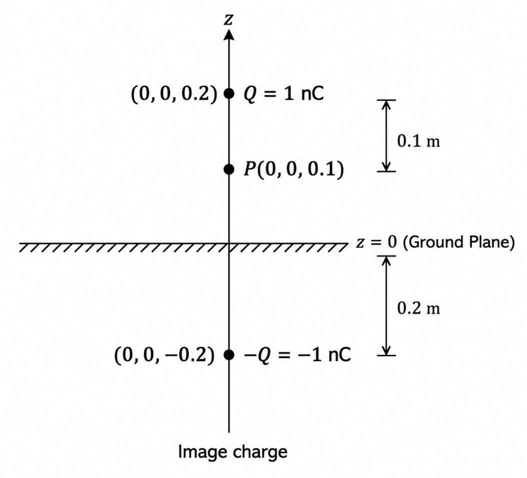

Fig. Solution Q1.18.2

The xy-plane ($z = 0$) is a conducting ground plane. Using the method of images, the image charge is

$$-q = -1\,\text{nC}, \qquad \text{located at } (0,0,\text{-}0.2)$$

We need the electric field at

$$P(0, 0, 0.1)$$

Field due to the real charge

Distance from the real charge to P:

$$R_1 = 0.2 - 0.1 = 0.1\,\text{m}$$

Hence,

$$E_1 = \frac{1}{4\pi\varepsilon_0}\frac{q}{R_1^2} = 8.99 \times 10^9 \cdot \frac{10^{-9}}{(0.1)^2} = 899.1\,\text{V/m}$$

Since P lies below the positive charge, the field is along $-\hat{z}$:

$$\vec{E}_1 = -899.1\,\hat{z}$$

Field due to the image charge

Distance from the image charge to P:

$$R_2 = 0.1 - (-0.2) = 0.3\,\text{m}$$

Hence,

$$E_2 = \frac{1}{4\pi\varepsilon_0}\frac{|q|}{R_2^2} = 8.99 \times 10^9 \cdot \frac{10^{-9}}{(0.3)^2} = 99.9\,\text{V/m}$$

Since the image charge is negative, the field at P is directed towards it, i.e. along $-\hat{z}$:

$$\vec{E}_2 = -99.9\,\hat{z}$$

Total electric field

$$\vec{E} = \vec{E}_1 + \vec{E}_2 = -(899.1 + 99.9)\hat{z}$$

$$\vec{E} = -999.0\,\hat{z}$$

Therefore,

$$\boxed{D}$$

Key Points

- The image charge is not real. The conductor actually develops an induced surface charge distribution. A positive charge above a grounded conducting plane induces negative charge on the conductor. Instead of solving for this complicated induced charge distribution, the method of images replaces it with an equivalent image charge that produces the same field in the region of interest.

Mistakes to be avoided

- Be careful with the direction of the electric field due to both the real charge and the image charge. The field due to a negative image charge is directed towards the image charge.
- Do not solve the problem by considering only the real charge. Since a conducting plane is present, its effect must be accounted for using the induced charges (or equivalently, the method of images).

Q1.17.1 NAT 2017 2M Ans: 10.03

Given,

$$R = 1\text{ cm} = 10^{-2}\text{ m}, \qquad a = 3.3\times 10^{-6}\text{ m}, \qquad V_1 = 1\text{ V}$$

For the soap bubble, the capacitance is

$$C_1 = 4\pi\varepsilon_0 R$$

Hence the charge on the bubble is

$$Q = C_1 V_1 = 4\pi\varepsilon_0 R$$

When the bubble bursts and forms a drop, charge is conserved:

$$Q_{\text{drop}} = Q$$

The volume of soap in the bubble is = $4\pi R^2 a$
and the volume of the spherical drop is,

$$\frac{4}{3}\pi r^3$$

So,

$$4\pi R^2 a = \frac{4}{3}\pi r^3 \Rightarrow r^3 = 3R^2 a$$

Substituting,

$$r^3 = 3(10^{-2})^2(3.3\times 10^{-6}) = 9.9\times 10^{-10}$$

$$r = (9.9\times 10^{-10})^{1/3} \approx 9.97\times 10^{-4}\text{ m}$$

Capacitance of the drop:

$$C_2 = 4\pi\varepsilon_0 r$$

Therefore,

$$V_2 = \frac{Q}{C_2} = \frac{4\pi\varepsilon_0 R}{4\pi\varepsilon_0 r} = \frac{R}{r}$$

$$V_2 = \frac{10^{-2}}{9.97\times 10^{-4}} \approx 10.03\text{ V}$$

$$\boxed{10.03}$$

Key Points

- If a soap bubble of radius R and potential V breaks into n identical drops, then by volume conservation:

$$r = \frac{R}{n^{1/3}}$$

Since charge is conserved,

$$V' = \frac{Q/n}{4\pi\varepsilon_0 r}$$

which gives

$$\boxed{V' = n^{1/3}V}$$

Q1.17.2 MCQ 2017 1M Ans: B

A perfect electric conductor (PEC) in electrostatic equilibrium has zero electric field inside it:

$$\vec{E} = 0$$

If any excess charge remained inside the conductor, it would produce an electric field within the conductor, which is not possible in electrostatic equilibrium.
Therefore, all excess charge resides on the outer surface of the conductor.
Since the conductor is spherical, every point on the surface is geometrically identical. Hence, the excess electrons distribute themselves uniformly over the outer surface.
Hence,

$$\boxed{B}$$

Q1.17.3 MCQ 2017 1M Ans: C

Consider the nature of each vector field.

Field X
The vectors radiate outward from a central point, indicating source/sink behaviour.
Hence, the net flux through a closed surface is non-zero and

$$\nabla \cdot X \neq 0$$

Since there is no circulation of field lines,

$$\nabla \times X = 0$$

Field Y
The vectors form closed circular loops.
Therefore, there is no source or sink and

$$\nabla \cdot Y = 0$$

However, the field possesses circulation, so

$$\nabla \times Y \neq 0$$

Field Z
The field lines spiral outward.
Hence, it exhibits both source/sink behaviour and circulation.
Therefore,

$$\nabla \cdot Z \neq 0$$

and

$$\nabla \times Z \neq 0$$

which matches

$$\boxed{C}$$

Q1.17.4 MCQ 2017 1M Ans: D

Let the initial positions of the particles be

$$x_e = -a, \qquad x_n = 0, \qquad x_p = +a$$

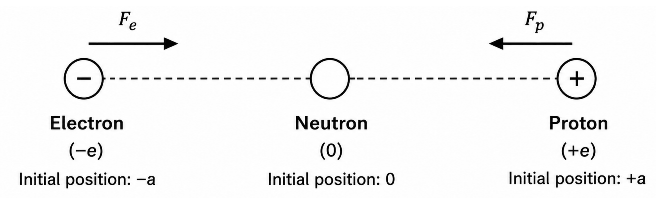

Fig. Solution Q1.17.4

The neutron is electrically neutral and therefore experiences no electrostatic force.
Hence, the neutron remains stationary at the origin.
The electron and proton attract each other due to the Coulomb force

$$F = \frac{1}{4\pi\varepsilon_0}\frac{e^2}{(2a)^2}$$

The magnitude of force on the electron and proton is the same.
Therefore accelaration,

$$a_e = \frac{F}{m_e}, \qquad a_p = \frac{F}{m_p}$$

Since

$$m_p \approx 1836\, m_e$$

we have

$$a_e \gg a_p$$

Thus, the electron accelerates much faster than the proton.
As the electron and proton move towards each other, they must pass through the midpoint $x = 0$, where the neutron is located.
Since the electron reaches the midpoint first, the first collision occurs between the electron and the neutron.
Therefore,

$$\boxed{D}$$

Q1.16.1 MCQ 2016 1M Ans: A

The two electrodes are maintained at the same potential.

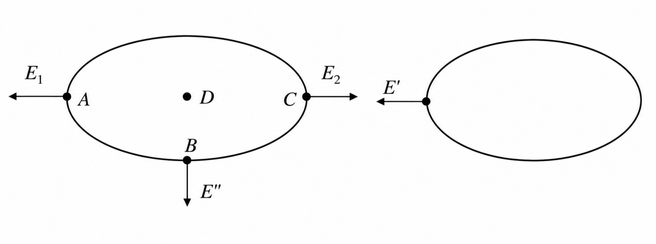

Fig. Solution Q1.16.1

Let E_1 be the field at point A due to the left electrode alone, and let E' be the field contribution due to the right electrode.
Point D lies inside the conductor. Hence, in electrostatic equilibrium,

$$E = 0$$

Therefore, point D cannot be the point of maximum electric field.
At point C, the field due to the left electrode is opposed by the field due to the right electrode. Hence,

$$E_C = E_2 - E'$$

Therefore, the net electric field at point C is reduced. Similarly, at point B, the net electric field is the vector sum of the two field contributions.
At point A, the fields due to the left and right electrodes are in the same direction. Hence,

$$E_A = E_1 + E'$$

Therefore, the net electric field is maximum at point A.

$$\boxed{\text{A}}$$

Key Points

- Maximum electric field is not always present at the point of minimum separation between conductors.
- The electric field inside a conductor in electrostatic equilibrium is zero.
- Electric field contributions must be added vectorially according to their directions.

Mistakes to be avoided

- Do not automatically choose point C merely because it lies in the smallest gap between the electrodes.
- Do not apply capacitor intuition involving opposite-potential electrodes; here both conductors are maintained at the same potential.
- Do not forget that the electric field inside a conductor in electrostatic equilibrium is zero.

Q1.16.2 MCQ 2016 1M Ans: C

Since both dielectric regions are connected across the same two plates, the potential difference across them is the same ($V_A = V_B$)

Also, the plate separation is identical ($d_A = d_B = 2$ cm)

The electric field between parallel plates is

$$E = \frac{V}{d}$$

Therefore,

$$E_A = \frac{V_A}{d_A}, \qquad E_B = \frac{V_B}{d_B}$$

Since

$$V_A = V_B \qquad \text{and} \qquad d_A = d_B$$

it follows that

$$E_A = E_B$$

Given,

$$E_A = 4 \text{ kV/cm} \qquad \Rightarrow \qquad E_B = 4 \text{ kV/cm}$$

Therefore,

$$\boxed{C}$$

Q1.16.3 MCQ 2016 2M Ans: D

Let the point on the zero equipotential curve be $P(x, y)$.

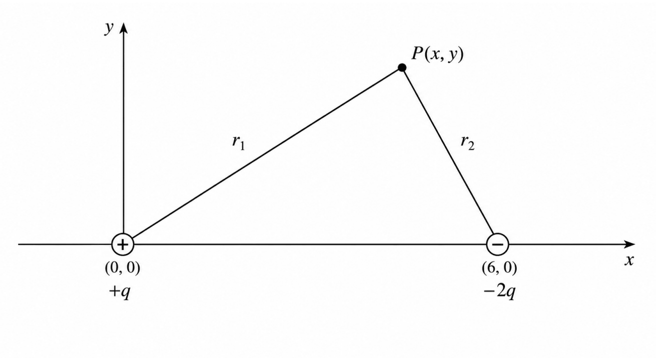

Fig. Solution Q1.16.3

Distance from charge q at $(0, 0)$:

$$r_1 = \sqrt{x^2 + y^2}$$

Distance from charge $-2q$ at $(6, 0)$:

$$r_2 = \sqrt{(x-6)^2 + y^2}$$

For zero potential,

$$\frac{kq}{r_1} - \frac{2kq}{r_2} = 0$$

Hence,

$$r_2 = 2r_1$$

Squaring both sides,

$$(x-6)^2 + y^2 = 4(x^2 + y^2)$$

$$3x^2 + 3y^2 + 12x - 36 = 0$$

Dividing by 3,

$$x^2 + y^2 + 4x - 12 = 0$$

Completing the square,

$$(x+2)^2 + y^2 = 16$$

Therefore,

$$\boxed{D}$$

Q1.16.4 MCQ 2016 1M Ans: B ● ○ ○

From Maxwell's equation,

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

The given ring is uniformly charged and is static.
A static charge distribution produces only an electrostatic field.
Since there is no time-varying magnetic field,

$$\frac{\partial \vec{B}}{\partial t} = 0$$

Hence,

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} = 0$$

Therefore, the curl of the electric field is zero.

$$\boxed{B}$$

Key Points

- Maxwell–Faraday equation:

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

- Electrostatic fields are conservative.
- For static charge distributions,

$$\frac{\partial \vec{B}}{\partial t} = 0$$

and therefore

$$\nabla \times \vec{E} = 0.$$

Mistakes to be avoided

- Do not confuse electrostatic fields with time-varying electromagnetic fields.
- A static electric field may have non-zero divergence due to charge, but its curl is always zero.
- Apply Maxwell–Faraday law only after checking whether the magnetic field varies with time.

Q1.15.1 MCQ | 2015 | 1M | Ans: A | ● ○ ○

Boundary conditions in wave propagation are

$$E_{\|1} = E_{\|2}$$

This means that the component of electric field parallel to the boundary of dielectric media remains same. Also,

$$\varepsilon_1 E_{\perp 1} = \varepsilon_2 E_{\perp 2}$$

This means that the perpendicular component of electric flux density remains the same.

Since the planes are separated by the boundary $y = 0$, the perpendicular component of the field is the y-component, and the parallel components are the x and z components.

Therefore,

$$E_{\|2} = 3\hat{a}_x + 2\hat{a}_z$$

For the normal component,

$$\varepsilon_1 E_{\perp 1} = \varepsilon_2 E_{\perp 2}$$

$$E_{\perp 2} = \frac{\varepsilon_1}{\varepsilon_2} E_{\perp 1}$$

$$E_{\perp 2} = \frac{2}{5}\left(4\hat{a}_y\right)$$

$$E_{\perp 2} = 1.6\hat{a}_y$$

Hence,

$$\vec{E}_2 = E_{\|2} + E_{\perp 2}$$

$$\vec{E}_2 = 3\hat{a}_x + 1.6\hat{a}_y + 2\hat{a}_z$$

Therefore, the correct option is

$$\boxed{A}$$

Key Points

- Tangential component of electric field is continuous at dielectric boundary: $E_{\|1} = E_{\|2}$.
- Normal component of electric flux density is continuous: $\varepsilon_1 E_{\perp 1} = \varepsilon_2 E_{\perp 2}$.
- For boundary $y = 0$, the normal component is along $\hat{a}_y$.
- Components parallel to boundary $y = 0$ are along $\hat{a}_x$ and $\hat{a}_z$.

Mistakes to be avoided

- Do not apply the same boundary condition to all components of electric field.
- Identify the normal direction correctly from the boundary equation.
- Do not change tangential components while crossing dielectric boundary.
- Use permittivity ratio only for the normal component of electric field.

Q1.15.2 MCQ 2015 2M Ans: D

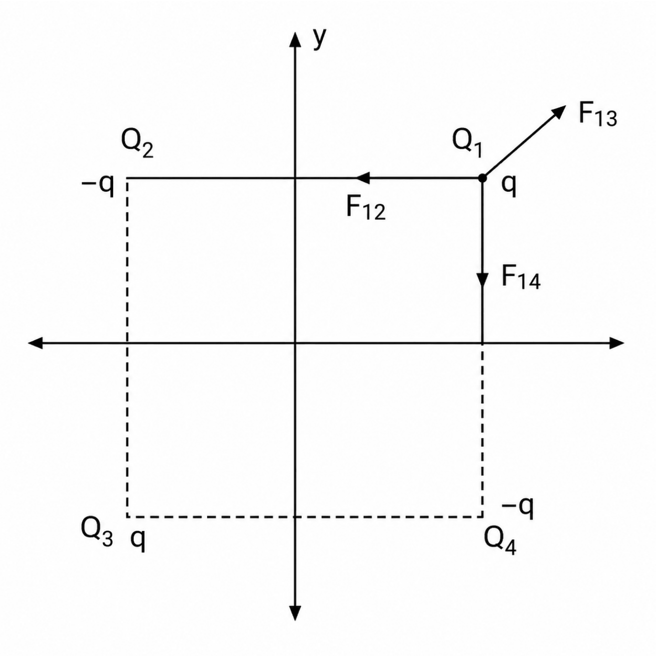

Fig. Solution Q1.15.2

By the image charge method, the charge configuration is replaced by suitable image charges and the force on charge Q is obtained by superposition.
The force components are

$$\vec{F}_{14} = \frac{Q^2}{4\pi\varepsilon_0(2d)^2}(-\hat{y})$$

$$\vec{F}_{12} = \frac{Q^2}{4\pi\varepsilon_0(2d)^2}(-\hat{x})$$

The diagonal image charge is at a distance

$$2\sqrt{2}\,d$$

Hence,

$$\vec{F}_{13} = \frac{Q^2}{4\pi\varepsilon_0(2\sqrt{2}d)^2}\left(\frac{\hat{x}}{\sqrt{2}} + \frac{\hat{y}}{\sqrt{2}}\right)$$

The net force is

$$\vec{F}_1 = \vec{F}_{12} + \vec{F}_{13} + \vec{F}_{14}$$

$$\vec{F}_1 = \frac{Q^2}{4\pi\varepsilon_0 d^2}\left[\left(\frac{1}{8\sqrt{2}} - \frac{1}{4}\right)\hat{x} + \left(\frac{1}{8\sqrt{2}} - \frac{1}{4}\right)\hat{y}\right]$$

Let

$$\vec{F}_1 = \frac{Q^2}{4\pi\varepsilon_0 d^2}\vec{K}$$

Therefore,

$$\vec{K} = \left(\frac{1}{8\sqrt{2}} - \frac{1}{4}\right)\hat{x} + \left(\frac{1}{8\sqrt{2}} - \frac{1}{4}\right)\hat{y}$$

$$\vec{K} = \frac{1-2\sqrt{2}}{8\sqrt{2}}\hat{x} + \frac{1-2\sqrt{2}}{8\sqrt{2}}\hat{y}$$

$$\vec{K} = \frac{1-2\sqrt{2}}{8\sqrt{2}}\hat{i} + \frac{1-2\sqrt{2}}{8\sqrt{2}}\hat{j}$$

Therefore, the correct option is

$$\boxed{\text{D}}$$

Key Points

- Image charge method replaces conducting boundaries with equivalent image charges.
- Net force is obtained by vector addition of forces due to all image charges.
- Coulomb force magnitude is inversely proportional to the square of separation distance.
- Diagonal image charge contributes equally along $\hat{x}$ and $\hat{y}$ directions.

Mistakes to be avoided

- Do not forget that the diagonal image charge is located at a distance $2\sqrt{2}\,d$.
- Resolve the diagonal force into $\hat{x}$ and $\hat{y}$ components using $1/\sqrt{2}$.
- Maintain proper signs while adding vector components.
- Do not add force magnitudes directly; perform vector addition.

Q1.15.3 NAT 2015 2M Ans: 18.75 ● ● ○

Capacitance of the air capacitor is

$$C_{\text{air}} = (\varepsilon_0 \times 1)\frac{A}{5\,\text{mm}} = C_1$$

Capacitance of the glass capacitor is

$$C_{\text{glass}} = (\varepsilon_0 \times 4)\frac{A}{5\,\text{mm}} = 4C_1$$

If air reaches its dielectric strength of 30 kV/cm, the maximum voltage across the air layer is

$$V_a = 30 \times 0.5 = 15\,\text{kV}$$

Since

$$X_C \propto \frac{1}{C},$$

the glass capacitor has one-fourth the reactance of the air capacitor.
Hence,

$$V_g = \frac{V_a}{4} = \frac{15}{4} = 3.75\,\text{kV}$$

Therefore, the total applied voltage is

$$V = V_a + V_g = 15 + 3.75 = 18.75\,\text{kV}$$

Although the dielectric strength of glass is 300 kV/cm, the air layer reaches its breakdown limit first. Therefore, dielectric breakdown occurs in the air before the glass reaches its breakdown voltage.

Hence, the correct answer is

$$\boxed{18.75}$$

Key Points

- Parallel-plate capacitance:

$$C = \varepsilon \frac{A}{d}.$$

- For capacitors in series, voltage divides inversely proportional to capacitance.
- Breakdown occurs first in the dielectric having the lower withstand voltage.

Mistakes to be avoided

- Do not assume equal voltage across dielectric layers connected in series.
- Do not compare only dielectric strengths; compare the actual voltage across each dielectric.
- Remember that the dielectric with larger capacitance has the smaller voltage drop.

Q1.14.1 MCQ 2014 1M Ans: A ● ○ ○

For a completely air-filled parallel plate capacitor,

$$C_0 = \frac{\varepsilon_0 A}{d}$$

A capacitor with half of its area filled with dielectric can be treated as two capacitors connected in parallel.

For the air-filled half,

$$C_1 = \frac{\varepsilon_0 \left(\frac{A}{2}\right)}{d} = \frac{\varepsilon_0 A}{2d}$$

For the dielectric-filled half,

$$C_2 = \frac{\varepsilon_0 \varepsilon_r \left(\frac{A}{2}\right)}{d} = \frac{\varepsilon_0 \varepsilon_r A}{2d}$$

Since the two capacitors are in parallel,

$$C_{\text{eq}} = C_1 + C_2$$

$$C_{\text{eq}} = \frac{\varepsilon_0 A}{2d} + \frac{\varepsilon_0 \varepsilon_r A}{2d}$$

$$C_{\text{eq}} = \frac{\varepsilon_0 A}{2d} (1 + \varepsilon_r)$$

Using

$$C_0 = \frac{\varepsilon_0 A}{d},$$

we get

$$C_{\text{eq}} = \frac{C_0}{2} (1 + \varepsilon_r)$$

Hence, the correct option is

$$\boxed{A}$$

Key Points

- Capacitance of a parallel plate capacitor is
$$C = \frac{\varepsilon A}{d}.$$
- If dielectric occupies half the plate area, the arrangement is equivalent to two capacitors in parallel.

Mistakes to be avoided

- Do not treat the arrangement as a series combination.
- Use half of the plate area, not half of the plate separation.
- Apply dielectric constant only to the dielectric-filled portion.

Q1.14.2 MCQ 2014 1M Ans: C ● ● ○

The arrangement can be considered a series combination of three capacitors.

Capacitance of the first dielectric layer is

$$C_1 = \frac{\varepsilon_0 \varepsilon_1 A}{d/4} = \frac{4A\varepsilon_0\varepsilon_1}{d}$$

The capacitance of the middle dielectric layer is

$$C_2 = \frac{\varepsilon_0 \varepsilon_2 A}{d/2} = \frac{2A\varepsilon_0\varepsilon_2}{d}$$

Capacitance of the third dielectric layer is

$$C_3 = \frac{\varepsilon_0 \varepsilon_1 A}{d/4} = \frac{4A\varepsilon_0\varepsilon_1}{d}$$

Since

$$C_1 = C_3,$$

the voltage across each of them is

$$V_{C_1} = V_{C_3} = 2\,\mathrm{V}.$$

The equivalent capacitance of C_1 and C_3 in series is

$$C_{eq} = \frac{C_1 C_3}{C_1 + C_3} = \frac{2A\varepsilon_0\varepsilon_1}{d}.$$

Hence, the voltage across C_{eq} is

$$V_{eq} = 2 + 2 = 4\,\mathrm{V}.$$

Therefore,

$$Q_{eq} = C_{eq} V_{eq} = \left(\frac{2A\varepsilon_0\varepsilon_1}{d}\right) 4.$$

The voltage across C_2 is

$$V_{C_2} = 10 - 4 = 6\,\mathrm{V}.$$

Hence,

$$Q = C_2 V_{C_2} = \left(\frac{2A\varepsilon_0\varepsilon_2}{d}\right) 6.$$

Since the capacitors are in series,

$$Q = Q_{\text{eq}}.$$

Therefore,

$$\left(\frac{2A\varepsilon_0\varepsilon_2}{d}\right) 6 = \left(\frac{2A\varepsilon_0\varepsilon_1}{d}\right) 4.$$

Thus,

$$\frac{\varepsilon_1}{\varepsilon_2} = \frac{3}{2}$$

Hence, the correct option is

$$\boxed{C}$$

Key Points

- Capacitance of a parallel-plate capacitor is
$$C = \frac{\varepsilon A}{d}.$$
- Capacitance is inversely proportional to plate separation.
- Capacitors connected in series carry the same charge.
- Voltage divides inversely proportional to capacitance in a series combination.

Q1.14.3 MCQ 2014 1M Ans: A ● ○ ○

Given,

$$\nabla \cdot (f\vec{v}) = x^2y + y^2z + z^2x$$

Using the vector identity,

$$\nabla \cdot (f\vec{v}) = f(\nabla \cdot \vec{v}) + \vec{v} \cdot \nabla f$$

The vector field is

$$\vec{v} = y\hat{i} + z\hat{j} + x\hat{k}$$

Therefore,

$$\nabla \cdot \vec{v} = \frac{\partial y}{\partial x} + \frac{\partial z}{\partial y} + \frac{\partial x}{\partial z} = 0$$

Hence,

$$\nabla \cdot (f\vec{v}) = \vec{v} \cdot \nabla f$$

So,

$$\vec{v} \cdot \nabla f = x^2y + y^2z + z^2x$$

Therefore,

$$\boxed{x^2y + y^2z + z^2x}$$

Hence, the correct option is

$$\boxed{A}$$

Key Points

- Product divergence identity:
$$\nabla \cdot (f\vec{v}) = f(\nabla \cdot \vec{v}) + \vec{v} \cdot \nabla f.$$
- If $\nabla \cdot \vec{v} = 0$, then $\nabla \cdot (f\vec{v}) = \vec{v} \cdot \nabla f$.
- For $\vec{v} = y\hat{i} + z\hat{j} + x\hat{k}$, divergence is zero.

Mistakes to be avoided

- Do not directly differentiate $x^2y + y^2z + z^2x$.
- First use the identity for $\nabla \cdot (f\vec{v})$.
- Do not forget that $\dfrac{\partial y}{\partial x} = \dfrac{\partial z}{\partial y} = \dfrac{\partial x}{\partial z} = 0$.

Q1.14.4 MCQ 2014 2M Ans: B ● ● ●

Since an infinite conducting plane is present, the problem is solved using the **method of images**. Replace the conducting plane by an image charge.

Let the observation point be

$$P(\sqrt{2},\ \sqrt{2},\ 0).$$

The real charge is

$$Q = 32\pi\varepsilon_0\sqrt{2}\,\mathrm{C}$$

located at

$$Q_1(0, 0, 2),$$

and the image charge is

$$-Q = -32\pi\varepsilon_0\sqrt{2}\,\mathrm{C}$$

located at

$$Q_2(0, 0, -2).$$

Distance from both charges to point P is

$$d_{P1} = \sqrt{(\sqrt{2}-0)^2 + (\sqrt{2}-0)^2 + (0-2)^2} = 2\sqrt{2},$$

$$d_{P2} = \sqrt{(\sqrt{2}-0)^2 + (\sqrt{2}-0)^2 + (0+2)^2} = 2\sqrt{2}.$$

The corresponding unit vectors are

$$\hat{d}_{P1} = \frac{\sqrt{2}\hat{i} + \sqrt{2}\hat{j} - 2\hat{k}}{2\sqrt{2}},$$

$$\hat{d}_{P2} = \frac{\sqrt{2}\hat{i} + \sqrt{2}\hat{j} + 2\hat{k}}{2\sqrt{2}}.$$

Electric field due to the real charge is

$$\vec{E}_{P1} = \frac{Q}{4\pi\varepsilon_0 d_{P1}^2}\hat{d}_{P1} = \sqrt{2}\left(\frac{\sqrt{2}\hat{i} + \sqrt{2}\hat{j} - 2\hat{k}}{2\sqrt{2}}\right)\text{N/C.}$$

Electric field due to the image charge is

$$\vec{E}_{P2} = \frac{-Q}{4\pi\varepsilon_0 d_{P2}^2}\hat{d}_{P2} = -\sqrt{2}\left(\frac{\sqrt{2}\hat{i} + \sqrt{2}\hat{j} + 2\hat{k}}{2\sqrt{2}}\right)\text{N/C.}$$

Hence,

$$\begin{aligned}\vec{E}_P &= \vec{E}_{P1} + \vec{E}_{P2} \\ &= \sqrt{2}\left(\frac{\sqrt{2}\hat{i} + \sqrt{2}\hat{j} - 2\hat{k}}{2\sqrt{2}}\right) - \sqrt{2}\left(\frac{\sqrt{2}\hat{i} + \sqrt{2}\hat{j} + 2\hat{k}}{2\sqrt{2}}\right) \\ &= -2\hat{k}\ \text{N/C.}\end{aligned}$$

Hence, the correct option is

$$\boxed{B}$$

Key Points

- For an infinite conducting plane, replace the conductor with an image charge of equal magnitude and opposite sign.
- Electric field due to a point charge is $\vec{E} = \frac{Q}{4\pi\varepsilon_0 r^2}\hat{r}$.
- The total electric field is the vector sum of the fields due to the real and image charges.

Q1.14.5 MCQ 2014 1M Ans: A ● ○ ○

The potential of a charged sphere is

$$V = \frac{Q}{4\pi\varepsilon_0 r}$$

Given,

$$V = 1\text{ V}$$

Therefore,

$$1 = \frac{Q}{4\pi\varepsilon_0 r}$$

$$Q = 4\pi\varepsilon_0 r$$

By Gauss's law,

$$\text{Flux} = \oint \vec{D}\cdot d\vec{A} = Q_{\text{enc}}$$

Hence,

$$\text{Flux} = 4\pi\varepsilon_0 r$$

Therefore, the correct option is

$$\boxed{\text{A}}$$

Key Points

- Potential of a charged conducting sphere is $V = \dfrac{Q}{4\pi\varepsilon_0 r}$.
- Electric flux density form of Gauss's law is $\oint \vec{D} \cdot d\vec{A} = Q_{\text{enc}}$.
- Once charge Q is obtained, total electric flux is equal to enclosed charge.

Q1.13.1 MCQ 2013 1M Ans: D

The given scalar field is

$$V = 2x^2y + 3y^2z + 4z^2x$$

For any scalar field V, the curl of its gradient is always zero.

$$\nabla \times (\nabla V) = 0$$

Therefore,

$$\nabla \times (\nabla V) = \vec{0}$$

Hence, the correct option is

$$\boxed{D}$$

Key Points

- Gradient of a scalar field gives a conservative vector field.
- Curl of a conservative vector field is always zero.
- Vector identity:
$$\nabla \times (\nabla V) = 0.$$

Q1.13.2 MCQ 2013 1M Ans: B

Given vector field is

$$\vec{F} = y^2x\,\hat{a}_x - yz\,\hat{a}_y - x^2\,\hat{a}_z$$

The line integral is evaluated along a segment on the x-axis from $x = 1$ to $x = 2$.
On the x-axis,

$$y = 0, \qquad z = 0$$

Also,

$$d\vec{l} = dx\,\hat{a}_x$$

Therefore,

$$\vec{F} \cdot d\vec{l} = \left(y^2x\right) dx$$

Since $y = 0$ on the x-axis,

$$\vec{F} \cdot d\vec{l} = 0$$

Hence,

$$\int_{x=1}^{x=2} \vec{F} \cdot d\vec{l} = \int_{1}^{2} 0\,dx = 0$$

Therefore, the correct option is

$$\boxed{B}$$

Q1.13.3 MCQ 2013 2M Ans: C ● ● ○

Let C_3 have its maximum voltage across it, i.e.,

$$V_3 = 2\,\text{V}$$

Then, the charge across C_3 is

$$Q_3 = C_3 V_3$$

$$Q_3 = 2 \times 2 = 4\,\mu\text{C}$$

Since C_2 and C_3 are connected in series, the charge across them will be same.

$$Q_3 = Q_2$$

$$Q_2 = C_2 V_2$$

$$4 = 5V_2$$

$$V_2 = 0.8\,\text{V}$$

Therefore, total voltage across the series path is

$$V_2 + V_3 = 0.8 + 2 = 2.8\,\text{V}$$

This voltage is also across the parallel branch containing C_1.
Hence, the total charge is

$$Q_{\text{total}} = Q_1 + Q_2$$

$$Q_{\text{total}} = C_1 V_1 + Q_2$$

$$Q_{\text{total}} = (10 \times 2.8) + 4$$

$$Q_{\text{total}} = 28 + 4 = 32\,\mu\text{C}$$

Hence, the correct option is

$$\boxed{C}$$

Key Points

- In series capacitors, charge remains same.
- In parallel branches, voltage remains same.
- Charge stored in a capacitor is $Q = CV$.

- Maximum voltage condition should be applied to the limiting capacitor first.

Mistakes to be avoided

- Do not assume same voltage across series capacitors.
- Do not add capacitances directly for the series branch.
- Use the voltage across the series branch as the voltage across the parallel capacitor.
- Keep units consistent: if C is in μF and V is in volts, then Q is in μC.

Q1.13.4 MCQ 2013 2M Ans: C

Energy density stored in an electric field is

$$u = \frac{1}{2}\varepsilon|\vec{E}|^2$$

Energy stored in the dielectric is

$$U = u \times \text{Volume}$$

Substituting the given values,

$$U = \frac{1}{2}\varepsilon_0\varepsilon_r\left(6^2 + 8^2\right) \times 10^{12} \times 0.5 \times 0.5 \times 0.4$$

$$U = \frac{1}{2} \times 8.85 \times 10^{-12} \times 2 \times 100 \times 10^{12} \times 0.1$$

$$U = 88.5\,\text{J}$$

Hence, the correct option is

$$\boxed{C}$$

Q1.11.1 MCQ 2011 2M Ans: C

Charge stored in a capacitor is

$$Q = CV$$

For a parallel plate capacitor,

$$C = \frac{\varepsilon_0\varepsilon_r A}{d}$$

Therefore,

$$Q = \frac{\varepsilon_0\varepsilon_r A}{d}V$$

Since,

$$E = \frac{V}{d},$$

we get

$$Q = \varepsilon_0\varepsilon_r AE$$

Substituting the given values,

$$Q = 8.85 \times 10^{-12} \times 2.26 \times 20 \times 10^{-2} \times 40 \times 10^{-2} \times 5000 \times 10^{3}$$

$$Q = 8\,\mu\text{C}$$

Hence, the correct option is

$$\boxed{C}$$

Q1.10.1 MCQ 2010 1M Ans: A ● ○ ○

The three-dimensional radial vector field is

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$$

Divergence is given by

$$\nabla \cdot \vec{r} = \frac{\partial x}{\partial x} + \frac{\partial y}{\partial y} + \frac{\partial z}{\partial z}$$

$$\nabla \cdot \vec{r} = 1 + 1 + 1 = 3$$

Hence, the correct option is

$$\boxed{A}$$

Q1.08.1 MCQ 2008 2M Ans: B ● ○ ○

Since the multiple dielectrics are divided along the thickness of the capacitor, they can be considered as two capacitors connected in series.

Let C_1 be the capacitance with glass dielectric and C_2 be the capacitance with paper dielectric.

$$C_{\text{eq}} = \frac{C_1 C_2}{C_1 + C_2}$$

where

$$C_1 = \frac{A\varepsilon_0\varepsilon_{r1}}{d_1}, \qquad C_2 = \frac{A\varepsilon_0\varepsilon_{r2}}{d_2}$$

Substituting into the series capacitance formula,

$$C = \frac{\left(\frac{A\varepsilon_0\varepsilon_{r1}}{d_1}\right)\left(\frac{A\varepsilon_0\varepsilon_{r2}}{d_2}\right)}{\frac{A\varepsilon_0\varepsilon_{r1}}{d_1} + \frac{A\varepsilon_0\varepsilon_{r2}}{d_2}}$$

which simplifies to

$$C = \frac{A\varepsilon_0\varepsilon_{r1}\varepsilon_{r2}}{\varepsilon_{r1}d_2 + \varepsilon_{r2}d_1}$$

Substituting the given values,

$$C = \frac{500 \times 500 \times 10^{-6} \times 8.85 \times 10^{-12} \times 8 \times 2}{(8 \times 2 + 2 \times 4) \times 10^{-3}}$$

$$C = 1475\,\text{pF}$$

$$\boxed{B}$$

Key Points

- Dielectrics stacked along the thickness are equivalent to capacitors connected in series.
- Capacitance of each dielectric layer is $C = \dfrac{\varepsilon_0 \varepsilon_r A}{d}$.
- For series capacitors,

$$C_{\text{eq}} = \frac{C_1 C_2}{C_1 + C_2}.$$

Q1.08.2 | MCQ | 2008 | 2M | Ans: C

Construct an imaginary Gaussian surface of cuboidal shape with the xy-plane at the center and height 40 units. Let the dimensions along x and y directions be infinite.

Then half of the flux passes through the plane $z = 20$ and the other half passes through the plane $z = -20$.
There is no flux through the planes perpendicular to x and y directions, as they are at infinite distance.
By Gauss's law, flux through a surface is equal to the charge enclosed by the Gaussian surface.
Therefore,

$$Q = \frac{Q_1 + Q_2}{2}$$

Substituting,

$$Q = \frac{10 + 20}{2}$$

$$Q = 15\,\mu\text{C}$$

Hence, the correct option is

$$\boxed{C}$$

Key Points

- According to Gauss's law, total electric flux through a closed surface is equal to the charge enclosed.
- For a symmetric Gaussian surface, flux divides equally through the two opposite faces.
- If the side faces are taken at infinite distance, flux through them is zero.

Q1.07.1 | MCQ | 2007 | 2M | Ans: A

By Gauss's law,

$$\oint \vec{E} \cdot d\vec{S} = \frac{Q_{\text{enc}}}{\varepsilon_0}$$

The volume charge density is

$$\rho = \frac{Q}{\frac{4}{3}\pi R^3}$$

The charge enclosed within a Gaussian sphere of radius r is

$$Q_{\text{enc}} = \rho\left(\frac{4}{3}\pi r^3\right) = \frac{Q}{\frac{4}{3}\pi R^3}\left(\frac{4}{3}\pi r^3\right) = Q\left(\frac{r}{R}\right)^3$$

Applying Gauss's law,

$$E(4\pi r^2) = \frac{Q_{enc}}{\varepsilon_0} = \frac{Q}{\varepsilon_0}\left(\frac{r}{R}\right)^3$$

Hence,

$$E = \frac{Qr^3}{4\pi\varepsilon_0 R^3 r^2} = \frac{Qr}{4\pi\varepsilon_0 R^3}$$

Thus, the electric field inside the uniformly charged sphere is

$$\boxed{E = \frac{Qr}{4\pi\varepsilon_0 R^3}}$$

Hence, the correct option is

$$\boxed{A}$$

Key Points

- Gauss's law: $\oint \vec{E} \cdot d\vec{S} = \frac{Q_{enc}}{\varepsilon_0}$.
- For a uniformly charged sphere, $\rho = \frac{Q}{\frac{4}{3}\pi R^3}$.
- Inside a uniformly charged sphere, the electric field varies linearly with the radial distance, i.e., $E \propto r$.

Q1.07.2 MCQ 2007 1M Ans: A

Given,

$$\vec{V}(x, y, z) = -(x\cos xy + y)\,\hat{i} + (y\cos xy)\,\hat{j} + \left(\sin z^2 + x^2 + y^2\right)\hat{k}$$

Divergence of a vector field is

$$\nabla \cdot \vec{V} = \frac{\partial V_x}{\partial x} + \frac{\partial V_y}{\partial y} + \frac{\partial V_z}{\partial z}$$

Here,

$$V_x = -(x\cos xy + y), \qquad V_y = y\cos xy, \qquad V_z = \sin z^2 + x^2 + y^2$$

Therefore,

$$\frac{\partial V_x}{\partial x} = -\cos xy + xy\sin xy$$

$$\frac{\partial V_y}{\partial y} = \cos xy - xy\sin xy$$

$$\frac{\partial V_z}{\partial z} = 2z\cos z^2$$

Hence,

$$\nabla \cdot \vec{V} = 2z\cos z^2$$

$$\boxed{A}$$

Q1.05.1 | MCQ | 2005 | 1M | Ans: D | ● ○ ○

Using the vector identity,

$$\nabla \cdot \left(\nabla \times \vec{A}\right) = 0$$

for any continuously differentiable vector field $\vec{A}$.
Here,

$$\vec{A} = \vec{E}$$

Therefore,

$$\nabla \cdot \left(\nabla \times \vec{E}\right) = 0$$

Hence, the given expression is a scalar equal to zero.

$$\boxed{D}$$

Q1.04.1 | MCQ | 2004 | 2M | Ans: D | ● ○ ○

For two dielectric layers placed one after another between the capacitor plates, the individual capacitances are connected in series.

The equivalent capacitance is

$$C_{\text{eq}} = \frac{C_1 C_2}{C_1 + C_2}$$

where

$$C_1 = \frac{\varepsilon_{r1}\varepsilon_0 A}{d_1}, \qquad C_2 = \frac{\varepsilon_{r2}\varepsilon_0 A}{d_2}$$

Substituting,

$$C = \frac{\left(\frac{\varepsilon_{r1}\varepsilon_0 A}{d_1}\right)\left(\frac{\varepsilon_{r2}\varepsilon_0 A}{d_2}\right)}{\frac{\varepsilon_{r1}\varepsilon_0 A}{d_1} + \frac{\varepsilon_{r2}\varepsilon_0 A}{d_2}}$$

Now,

$$\varepsilon_0 = 8.85 \times 10^{-12}\,\text{F/m}, \quad A = 400 \times 10^{-3}\,\text{m}^2,$$

$$\varepsilon_{r1} = 4, \quad \varepsilon_{r2} = 2, \quad d_1 = 8 \times 10^{-3}\,\text{m}, \quad d_2 = 6 \times 10^{-3}\,\text{m}$$

Hence,

$$C = \frac{(8.85 \times 10^{-12})(400 \times 10^{-3}) \times 4 \times 2}{(4 \times 8 \times 10^{-3}) + (6 \times 2 \times 10^{-3})}$$

$$C = 257 \times 10^{-12}\,\text{F} = 257\,\text{pF}$$

$$\boxed{D}$$

Q1.03.1 MCQ 2003 2M Ans: B

The electric potential due to a point charge is

$$V = \frac{q}{4\pi\varepsilon_0 r}$$

where r is the distance of the point from the charge.
Hence, the potential difference between points P and Q is

$$V_P - V_Q = \frac{q}{4\pi\varepsilon_0(OP)} - \frac{q}{4\pi\varepsilon_0(OQ)}$$

Substituting,

$$V_P - V_Q = 9 \times 10^9 \left[\frac{10^{-9}}{40 \times 10^{-3}} - \frac{10^{-9}}{20 \times 10^{-3}}\right]$$

$$= 9\,(25 - 50)$$

$$= -225\ \text{V}$$

Therefore,

$$\boxed{B}$$

Key Points

- Electric potential due to a point charge is
$$V = \frac{q}{4\pi\varepsilon_0 r}.$$
- Potential is a scalar quantity and varies inversely with the distance from the charge.
- Potential difference is obtained by subtracting the potentials at the two points.

Mistakes to be avoided

- Do not confuse electric potential with electric field; potential varies as $\frac{1}{r}$, whereas the electric field varies as $\frac{1}{r^2}$.

Q1.03.2 MCQ 2003 2M Ans: D

The energy stored in a capacitor is

$$E = \frac{1}{2}QV = \frac{1}{2}CV^2$$

For a parallel-plate capacitor,

$$C = \frac{\varepsilon_0 A}{d}$$

Hence,

$$E = \frac{1}{2}\frac{\varepsilon_0 A}{d}V^2$$

Substituting,

$$\varepsilon_0 = 8.85 \times 10^{-12}\ \text{F/m}, \quad A = 100 \times 10^{-6}\ \text{m}^2,$$

$$d = 10^{-4}\,\text{m}, \quad V = 10^4\,\text{V}$$

$$E = \frac{1}{2} \times \frac{8.85 \times 10^{-12} \times 100 \times 10^{-6} \times (10^4)^2}{10^{-4}}$$

$$E = 44.3 \times 10^{-9}\,\text{J} = 44.3\,\text{nJ}$$

$$\boxed{D}$$

Key Points

- Energy stored in a capacitor is
$$E = \frac{1}{2}CV^2 = \frac{1}{2}QV = \frac{Q^2}{2C}.$$
- Capacitance of a parallel-plate capacitor is
$$C = \frac{\varepsilon_0 A}{d}.$$
- Energy stored increases directly with capacitance and is proportional to the square of the applied voltage.

Q1.03.3 | MCQ | 2003 | 2M | Ans: B | ● ○ ○

The given dielectric combination can be treated as two capacitors connected in series.

The individual capacitances are

$$C_1 = \frac{\varepsilon_{r1}\varepsilon_0 A}{t_1}$$

and

$$C_2 = \frac{\varepsilon_{r2}\varepsilon_0 A}{t_2}$$

Since the capacitors are connected in series, the charge on both capacitors is the same. Hence,

$$C_1(100 - V) = C_2 V$$

Substituting the expressions for C_1 and C_2,

$$\frac{\varepsilon_{r1}\varepsilon_0 A}{t_1}(100 - V) = \frac{\varepsilon_{r2}\varepsilon_0 A}{t_2}V$$

Given,

$$\varepsilon_{r1} = 6, \qquad t_1 = 4,$$

$$\varepsilon_{r2} = 4, \qquad t_2 = 2$$

Therefore,

$$\frac{6}{4}(100 - V) = \frac{4}{2}V$$

$$6(100 - V) = 8V$$

$$600 = 14V$$

$$V = \frac{600}{10} = 60\,\text{V}$$

$$\boxed{B}$$

Q1.01.1 MCQ | 2001 | 2M | Ans: A | ● ● ○

The position vector of the observation point is

$$\vec{r}_1 = \hat{i} + \hat{j} + 0\hat{k}$$

The position vector of the charge is

$$\vec{r}_2 = -\hat{i} + \hat{j} + \hat{k}$$

Hence, the position vector of the observation point with respect to the charge is

$$\vec{r} = \vec{r}_1 - \vec{r}_2 = 2\hat{i} - \hat{k}$$

The magnitude of $\vec{r}$ is

$$r = \sqrt{2^2 + (-1)^2} = \sqrt{5}$$

The electric field due to a point charge is

$$\vec{E} = \frac{Q}{4\pi\varepsilon_0 r^3}\vec{r}$$

Substituting,

$$\vec{E} = \frac{10^{-6}}{4\pi\varepsilon_0(\sqrt{5})^3}\left(2\hat{i} - \hat{k}\right)$$

Since,

$$(\sqrt{5})^3 = 5\sqrt{5},$$

we obtain

$$\boxed{\vec{E} = \frac{10^{-6}}{20\sqrt{5}\,\pi\varepsilon_0}\left(2\hat{i} - \hat{k}\right)}$$

Hence, the correct answer is

$$\boxed{A}$$

Key Points

- The displacement vector from the charge to the observation point is
$$\vec{r} = \vec{r}_1 - \vec{r}_2.$$
- The electric field due to a point charge is
$$\vec{E} = \frac{Q}{4\pi\varepsilon_0 r^3}\vec{r}.$$
- The direction of the electric field is along the displacement vector from the charge to the observation point.

Q1.01.2 NAT 2001 5M Ans: *

The electric field intensity is related to electric potential as

$$\vec{E} = -\nabla V$$

Therefore,

$$\vec{E} = -\frac{\partial V}{\partial x}\hat{i} - \frac{\partial V}{\partial y}\hat{j} - \frac{\partial V}{\partial z}\hat{k}$$

Given potential gives,

$$\vec{E} = -100x\hat{i} - 100y\hat{j} - 100z\hat{k} \text{ V/m}$$

At the point $(+1, -1, +1)$,

$$\vec{E} = -100\hat{i} + 100\hat{j} - 100\hat{k}$$

The magnitude of the electric field is

$$|\vec{E}| = \sqrt{(-100)^2 + (100)^2 + (-100)^2} = 100\sqrt{3} \text{ V/m}$$

Hence, the unit vector in the field direction is

$$\hat{a}_E = \frac{\vec{E}}{|\vec{E}|} = \frac{-100\hat{i} + 100\hat{j} - 100\hat{k}}{100\sqrt{3}}$$

$$\hat{a}_E = \frac{-\hat{i} + \hat{j} - \hat{k}}{\sqrt{3}}$$

None of the option is matching, so

$$\boxed{none}$$

Q1.99.1 MCQ 1999 5M Ans: *

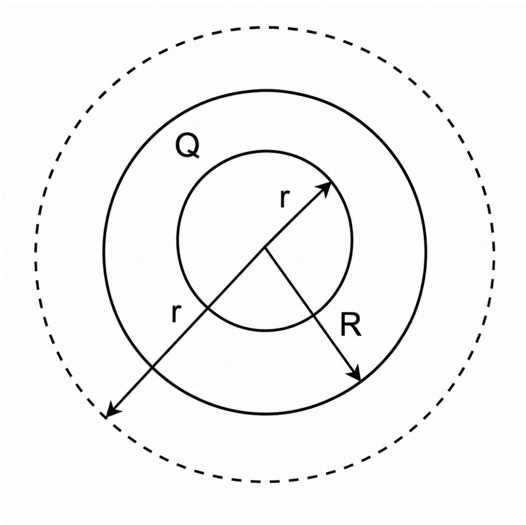

Fig. Solution Q1.99.1

For a uniformly charged solid sphere of radius R,

$$0 < r < R$$

the uniform volume charge density is

$$\rho_v = \frac{Q}{\frac{4}{3}\pi R^3}$$

The charge enclosed within a Gaussian sphere of radius r is

$$Q_{\text{enc}} = \rho_v \left(\frac{4}{3}\pi r^3\right) = Q\left(\frac{r}{R}\right)^3$$

Applying Gauss's law,

$$\oint \vec{D} \cdot d\vec{S} = Q_{\text{enc}}$$

$$D(4\pi r^2) = Q\left(\frac{r}{R}\right)^3$$

Hence,

$$D = \frac{Qr}{4\pi R^3}$$

Since

$$\vec{D} = \varepsilon \vec{E},$$

the electric field inside the sphere is

$$E = \frac{D}{\varepsilon} = \frac{Qr}{4\pi\varepsilon_0\varepsilon_r R^3}$$

At the surface,

$$r = R,$$

$$E_1 = \frac{Q}{4\pi\varepsilon_0\varepsilon_r R^2}$$

For the region outside the sphere,

$$r > R,$$

the enclosed charge is

$$Q_{\text{enc}} = Q$$

Again, by Gauss's law,

$$D(4\pi r^2) = Q$$

Therefore,

$$E = \frac{D}{\varepsilon_0} = \frac{Q}{4\pi\varepsilon_0 r^2}$$

At the surface,

$$r = R,$$

$$E_2 = \frac{Q}{4\pi\varepsilon_0 R^2}$$

Hence,

$$E_2 = \varepsilon_r E_1$$

which satisfies the discontinuity at the dielectric boundary.
The plot of electric field shown below.

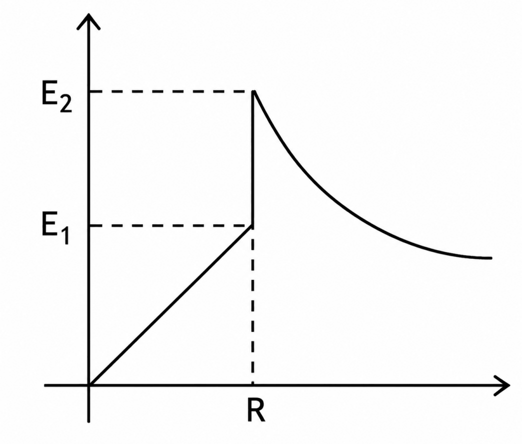

Fig. Solution Q1.99.1

Key Points

- For a uniformly charged solid sphere, the electric field inside varies linearly with radius:

$$E \propto r.$$

- Outside the sphere, the electric field behaves like that of a point charge:

$$E \propto \frac{1}{r^2}.$$

- At the dielectric-air interface,

$$E_{\text{outside}} = \varepsilon_r E_{\text{inside}}.$$

Mistakes to be avoided

- Do not use the total charge Q while applying Gauss's law for $r < R$; use only the enclosed charge.
- Use $\varepsilon = \varepsilon_0 \varepsilon_r$ inside the dielectric and $\varepsilon = \varepsilon_0$ outside.
- The electric field is discontinuous at the dielectric boundary, whereas the electric flux density $\vec{D}$ is continuous in the absence of free surface charge.

Q1.99.2 MCQ | 1999 | 1M | Ans: A | ● ○ ○

The electric field between the plates is

$$E = \frac{V}{d}$$

As long as the potential difference V and the separation d are kept constant, the electric field remains unchanged.
The capacitance of a parallel plate capacitor is

$$C = \frac{\varepsilon A}{d}$$

Hence, increasing the plate area A increases the capacitance.

Since

$$Q = CV,$$

and V is constant,

$$Q \propto C \propto A.$$

Therefore, the charge stored on the plates increases with the plate area. Since the electrostatic force on a plate is proportional to the charge for a constant electric field,

$$F = QE,$$

the force on the plate also increases.
Hence, the correct option is

$$\boxed{A}$$

Q1.98.1 MCQ 1998 5M Ans: * ● ● ○

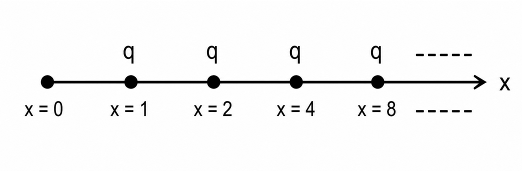

Fig. Solution Q1.98.1

The electric potential at $x = 0$ is the sum of the potentials due to all the charges.

$$V = \frac{q}{4\pi\varepsilon}\left(1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots\right)$$

Using the sum of the infinite geometric series,

$$1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots = \frac{1}{1 - \frac{1}{2}} = 2,$$

we get

$$V = \frac{q}{4\pi\varepsilon} \times 2 = \frac{q}{2\pi\varepsilon}\ \text{V}.$$

The electric field at $x = 0$ is directed along the negative x-axis due to all the positive charges lying on the positive x-axis.

$$\vec{E} = \frac{q}{4\pi\varepsilon}\left(1 + \frac{1}{2^2} + \frac{1}{4^2} + \cdots\right)(-\hat{a}_x)$$

Again, using the infinite geometric series,

$$1 + \frac{1}{2^2} + \frac{1}{4^2} + \cdots = \frac{1}{1 - \frac{1}{4}} = \frac{4}{3},$$

we obtain

$$\vec{E} = \frac{q}{4\pi\varepsilon}\left(\frac{4}{3}\right)(-\hat{a}_x) = -\frac{q}{3\pi\varepsilon}\,\hat{a}_x\ \text{V/m}.$$

Hence,

$$\boxed{V = \frac{q}{2\pi\varepsilon}\text{ V}, \qquad \vec{E} = -\frac{q}{3\pi\varepsilon}\hat{a}_x \text{ V/m}}$$

Q1.98.2 MCQ 1998 1M Ans: C

According to Faraday's law of electromagnetic induction, the induced emf is equal to the negative rate of change of magnetic flux linkage.

Mathematically,

$$e = -\frac{d\psi}{dt}$$

where ψ is the flux linkage.

The negative sign represents Lenz's law, which states that the induced emf always opposes the change in magnetic flux producing it.

Hence, the correct option is

$$\boxed{C}$$

Q1.97.1 MCQ 1997 1M Ans: D

In an electrostatic field, the electric field is always directed along the direction of maximum decrease of electric potential.

Since equipotential surfaces have the same potential everywhere, no work is done in moving a charge along an equipotential surface. Therefore, the electric field must be perpendicular to the equipotential surfaces (or equipotential lines in a two-dimensional representation).

Therefore, the correct option is

$$\boxed{D}$$

Q1.97.2 NAT 1997 2M Ans: 4.72

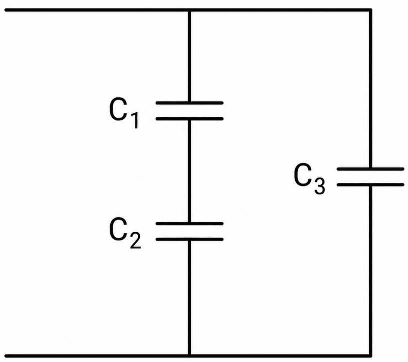

Fig. Solution Q1.97.2

The given dielectric arrangement can be treated as a combination of three capacitors.

The equivalent capacitance is

$$C = \frac{C_1 C_2}{C_1 + C_2} + C_3.$$

Here,

$$C_1 = \frac{\varepsilon_0 \varepsilon_1 \left(\frac{A}{2}\right)}{\frac{d}{2}}, \qquad C_2 = \frac{\varepsilon_0 \varepsilon_2 \left(\frac{A}{2}\right)}{\frac{d}{2}}, \qquad C_3 = \frac{\varepsilon_0 \varepsilon_3 \left(\frac{A}{2}\right)}{d}.$$

Substituting,

$$C = \frac{\left(\frac{\varepsilon_0 \varepsilon_1 A}{d}\right)\left(\frac{\varepsilon_0 \varepsilon_2 A}{d}\right)}{\frac{\varepsilon_0 \varepsilon_1 A}{d} + \frac{\varepsilon_0 \varepsilon_2 A}{d}} + \frac{\varepsilon_0 \varepsilon_3 A}{2d}.$$

Hence,

$$C = \frac{\varepsilon_0 A}{d}\left(\frac{\varepsilon_1 \varepsilon_2}{\varepsilon_1 + \varepsilon_2} + \frac{\varepsilon_3}{2}\right).$$

Substituting

$$\varepsilon_0 = 8.85 \times 10^{-12}\,\text{F/m}, \quad A = 400 \times 10^{-4}\,\text{m}^2, \quad d = 20 \times 10^{-2}\,\text{m},$$

$$\varepsilon_1 = 1, \qquad \varepsilon_2 = 2, \qquad \varepsilon_3 = 4,$$

gives

$$C = \frac{8.85 \times 10^{-12} \times 400 \times 10^{-4}}{20 \times 10^{-2}}\left[\frac{1 \times 2}{1 + 2} + \frac{4}{2}\right].$$

$$C = 4.72 \times 10^{-12}\,\text{F} = 4.72\,\text{pF}.$$

$$\boxed{4.72}$$

Q1.97.3 NAT | 1997 | 2M | Ans: 11.1 | ● ○ ○

The capacitance of an isolated conducting sphere of radius R in air is

$$C = 4\pi\varepsilon_0 R.$$

Given,

$$R = 10\,\text{cm} = 0.1\,\text{m},$$

and

$$4\pi\varepsilon_0 = \frac{1}{9 \times 10^9} = 1.111 \times 10^{-10}\,\text{F/m}.$$

Hence,

$$C = 4\pi\varepsilon_0(0.1) = \frac{0.1}{9 \times 10^9} = 1.11 \times 10^{-11}\,\text{F}.$$

Converting into picofarads,

$$C = 1.11 \times 10^{-11} \times 10^{12} = 11.1\,\text{pF}.$$

Therefore,

$$\boxed{11.1}$$

Key Points

- Capacitance of an isolated conducting sphere:

$$C = 4\pi\varepsilon_0 R.$$

- Capacitance depends only on the radius of the sphere and the surrounding medium.
- In air (or vacuum),

$$4\pi\varepsilon_0 = \frac{1}{9\times 10^9}\ \text{F/m}.$$

- 1 pF = 10^{-12} F.

Q1.97.4 MCQ 1997 5M Ans: *

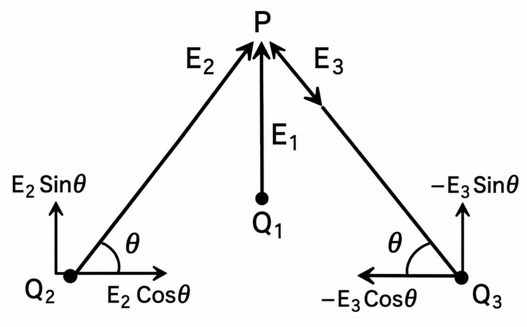

Fig. Solution Q1.97.4

Since $R \gg a$, the distance of point P from all the charges can be approximated as R.
The electric field due to charge Q_1 is

$$\vec{E}_1 = \frac{1}{4\pi\varepsilon_0}\frac{Q_1}{R^2}\hat{a}_y = 9\times 10^9 \times \frac{2000\times 10^{-12}}{1^2}\hat{a}_y = 18\,\hat{a}_y\ \text{V/m}.$$

Since the distances of Q_2 and Q_3 from P are equal and

$$|Q_2| = |Q_3|,$$

their magnitudes of electric field are equal.

$$|\vec{E}_2| = |\vec{E}_3| = \frac{1}{4\pi\varepsilon_0}\frac{Q_2}{R^2} = 9\times 10^9 \times \frac{1000\times 10^{-12}}{1^2} = 9\ \text{V/m}.$$

Since Q_3 is negative, the electric field at P is directed towards Q_3, whereas for the positive charge Q_2, it is directed away from the charge.
Hence, the vertical components of $\vec{E}_2$ and $\vec{E}_3$ cancel each other, while their horizontal components add.
From the geometry,

$$\cos\theta = \frac{B}{H} = \frac{a/2}{R} = \frac{1}{200}.$$

Therefore,

$$\vec{E} = \vec{E}_1 + \vec{E}_2 + \vec{E}_3 = E_1\hat{a}_y + 2E_2\cos\theta\,\hat{a}_x.$$

Substituting,

$$\vec{E} = 18\,\hat{a}_y + 2 \times 9 \times \frac{1}{200}\,\hat{a}_x = \left(0.09\,\hat{a}_x + 18\,\hat{a}_y\right)\ \text{V/m}.$$

Hence,

$$\boxed{\vec{E} = \left(0.09\,\hat{a}_x + 18\,\hat{a}_y\right)\ \text{V/m}}$$

Key Points

- For $R \gg a$, all source-to-observation distances may be approximated as R.
- Electric field due to a point charge is
$$\vec{E} = \frac{1}{4\pi\varepsilon_0}\frac{Q}{R^2}\,\hat{a}_R.$$
- Use symmetry to identify cancelling and reinforcing components.
- Resolve inclined electric fields into horizontal and vertical components before vector addition.

Mistakes to be avoided

- Do not forget that the field due to a negative charge is directed towards the charge.
- Do not add electric field magnitudes directly; perform vector addition.
- Use the approximation $R \gg a$ only after identifying the direction of each field.
- Ensure that
$$\cos\theta = \frac{a/2}{R}$$
is used for the horizontal component.

Q1.96.1 MCQ 1996 1M Ans: C

Electric potential difference is defined as the work done per unit test charge in bringing a test charge from one point to another.

Mathematically,

$$dV = \frac{dW}{dq}$$

where

- dW is the work done,
- dq is the test charge.

Hence, the correct option is

$$\boxed{C}$$

Q1.96.2 MCQ 1996 1M Ans: A

Inside a hollow conducting sphere under electrostatic equilibrium, all excess charges reside on the outer surface of the conductor.

Applying Gauss's law to any Gaussian surface entirely inside the hollow cavity,

$$\oint \vec{E}\cdot d\vec{A} = \frac{Q_{\text{enclosed}}}{\varepsilon_0} = 0.$$

Hence,

$$\vec{E} = 0.$$

Therefore, the correct option is

$$\boxed{A}$$

Key Points

- Electric field inside a conductor at electrostatic equilibrium is zero.
- All excess charge resides on the outer surface of the conductor.
- From Gauss's law,
$$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{\text{enclosed}}}{\varepsilon_0}.$$
- Since no charge is enclosed inside the cavity,
$$\vec{E} = 0.$$

Q1.95.1 MCQ 1995 1M Ans: D

The conducting shell is a perfect conductor. Hence, the electric field inside the conducting material is zero.

$$\vec{E} = 0$$

Since

$$\vec{E} = -\nabla V,$$

zero electric field means that the electric potential remains constant everywhere inside the conducting shell. Hence, the potential of the inner surface is equal to the potential at radius r.

$$V(r_1) = V(r)$$

The potential is therefore

$$V(r) = \frac{q}{4\pi\varepsilon_0 r_1}$$

Hence, the correct option is

$$\boxed{D}$$

Key Points

- Electric field inside a perfect conductor is zero.
- If $\vec{E} = 0$, then electric potential is constant.
- The entire conducting shell remains at the same potential.
- Potential due to charge q at radius r_1 is
$$V = \frac{q}{4\pi\varepsilon_0 r_1}.$$

Mistakes to be avoided

- Do not assume that zero electric field means zero potential.
- Inside a conductor, potential is constant, not necessarily zero.
- Do not use distance r if the shell potential is determined by radius r_1.

Q1.94.1 NAT 1994 1M Ans: V

A hollow conductor is an equipotential body. Hence, every point on its surface is at the same potential V. Since there is no charge inside the hollow region, the electric field is

$$\vec{E} = 0.$$

The electric field is related to the electric potential by

$$\vec{E} = -\frac{dV}{dr}.$$

Since

$$\vec{E} = 0,$$

it follows that

$$\frac{dV}{dr} = 0.$$

Hence, the potential gradient is zero, and the electric potential remains constant throughout the hollow conductor as well as inside the cavity.
Therefore, the potential difference between any point on the surface and any point inside the hollow conductor is zero.
Hence, the final answer is

$$\boxed{V}$$

Key Points

- A conductor in electrostatic equilibrium is an equipotential body.
- The electric field inside an empty cavity of a conductor is zero.
- Zero electric field implies
$$\frac{dV}{dr} = 0,$$
so the potential remains constant.

Q1.94.2 NAT 1994 1M Ans: True

An electrostatic field is a conservative field. Therefore, the closed line integral of the electrostatic field is always zero.

Applying Stokes' theorem,

$$\oint \vec{E} \cdot d\vec{l} = \iint (\nabla \times \vec{E}) \cdot d\vec{s}.$$

Since the electrostatic field is conservative,

$$\oint \vec{E} \cdot d\vec{l} = 0.$$

Hence,

$$\nabla \times \vec{E} = 0.$$

Therefore, the given statement is

$$\boxed{\text{True}}$$

Q1.94.3 MCQ 1994 1M Ans: C ● ○ ○

When a charge is given to a conductor, the free charges move until electrostatic equilibrium is established. At equilibrium, all excess charge resides on the outer surface of the conductor. The surface charge density is not uniform in general; it is greater at regions having a smaller radius of curvature. Hence, the surface charge density is inversely proportional to the radius of curvature.
Mathematically,

$$\sigma \propto \frac{1}{R},$$

where R is the local radius of curvature.
Therefore, the correct option is

$$\boxed{\text{C}}$$

Key Points

- Excess charge on a conductor resides only on its outer surface.
- Surface charge density is maximum at sharp edges or pointed regions.
- Surface charge density varies inversely with the radius of curvature:

$$\sigma \propto \frac{1}{R}.$$

- At electrostatic equilibrium, the electric field inside a conductor is zero.

Mistakes to be avoided

- Do not assume that charge is distributed uniformly over the surface of an irregular conductor.
- Excess charge does not remain distributed throughout the volume of a conductor in electrostatic equilibrium.
- Do not confuse the local radius of curvature with the overall radius of the conductor.
- A smaller radius of curvature implies a higher surface charge density.

Q1.92.1 NAT 1992 1M Ans: 2.236 ● ○ ○

The given electric potential is

$$V = 2x\sqrt{y}.$$

The electric field is obtained from the negative gradient of the potential,

$$\vec{E} = -\nabla V.$$

Evaluating the partial derivatives,

$$\frac{\partial V}{\partial x} = 2\sqrt{y}, \qquad \frac{\partial V}{\partial y} = \frac{x}{\sqrt{y}}.$$

Hence,

$$\vec{E} = -2\sqrt{y}\,\hat{a}_x - \frac{x}{\sqrt{y}}\,\hat{a}_y.$$

At the point $(1, 1)$,

$$\vec{E} = -2\hat{a}_x - \hat{a}_y.$$

Therefore, the magnitude of the electric field is

$$|\vec{E}| = \sqrt{(-2)^2 + (-1)^2} = \sqrt{5}\text{ V/m}.$$

$$|\vec{E}| = 2.236\text{ V/m}.$$

Hence,

$$\boxed{2.236}$$

Key Points

- Electric field is the negative gradient of potential:

$$\vec{E} = -\nabla V.$$

- In Cartesian coordinates,

$$\nabla V = \frac{\partial V}{\partial x}\hat{a}_x + \frac{\partial V}{\partial y}\hat{a}_y + \frac{\partial V}{\partial z}\hat{a}_z.$$

- Magnitude of a vector:

$$|\vec{E}| = \sqrt{E_x^2 + E_y^2 + E_z^2}.$$

Q1.92.2 MSQ 1992 1M Ans: A,D

Gauss's law in integral form is

$$\oiint_S \vec{D} \cdot d\vec{s} = \iiint_V \rho\, dv.$$

Hence, option (A) is correct.
The differential form of Gauss's law is

$$\nabla \cdot \vec{D} = \rho.$$

For a homogeneous isotropic medium,

$$\vec{D} = \varepsilon \vec{E},$$

where ε is constant. Therefore,

$$\nabla \cdot (\varepsilon \vec{E}) = \rho$$

$$\varepsilon \nabla \cdot \vec{E} = \rho$$

$$\nabla \cdot \vec{E} = \frac{\rho}{\varepsilon}.$$

Hence, option (D) is also correct.
Option (B),

$$\nabla \times \vec{H} = \vec{J},$$

represents Ampère's law (magnetostatics).
Option (C),

$$\nabla \cdot \vec{J} + \frac{\partial \rho}{\partial t} = 0,$$

is the continuity equation. The given equation is incomplete and is not Gauss's law.
Therefore,

$$\boxed{A, D}$$

Q1.91.1 MCQ 1991 1M Ans: C

The damaged portion behaves like an air-filled void inside the dielectric.
The effective capacitance is only slightly affected because the damaged area is very small compared to the total plate area:

$$A = 120\,\text{mm}^2,$$

whereas the hole area is

$$A_h = \frac{\pi}{4}(0.5)^2 \approx 0.196\,\text{mm}^2,$$

which is only about

$$\frac{A_h}{A} \times 100 \approx 0.16\%.$$

Hence, the capacitance and stored charge remain almost unchanged.
However, the electric field inside the air-filled void is much higher than that in mica because the normal component of electric flux density is continuous:

$$D_n = \varepsilon_0 E_{\text{air}} = \varepsilon_0 \varepsilon_r E_{\text{mica}}.$$

Therefore,

$$E_{\text{air}} = \varepsilon_r E_{\text{mica}} = 7E_{\text{mica}}.$$

Since the dielectric strength of air is much lower than that of mica, breakdown is likely to occur first in the damaged region.
Hence, the property most significantly affected is the dielectric breakdown strength.

$$\boxed{\text{C}}$$

Key Points

- Small voids have negligible effect on capacitance because their area is very small.
- Electric flux density is continuous across dielectric interfaces:

$$D_n = \varepsilon E.$$

- Electric field inside an air void is

$$E_{\text{air}} = \varepsilon_r E_{\text{dielectric}}.$$

- Air has much lower dielectric strength than mica, making breakdown more likely.

Mistakes to be avoided

- Do not assume that a small defect significantly changes capacitance.
- Do not neglect electric field enhancement inside air voids.
- Remember that dielectric failure usually initiates at defects or voids where the electric field is concentrated.

DEBUG INFO

Chapter I

Electrostatics

Total Questions : 65

MCQ : 51

MSQ : 1

NAT : 13

SOLUTIONS

II

Magnetostatics

Topics

Q2.26.1 MCQ 2026 1M Ans: A ● ● ○

The loop lies in the XY plane. Hence, its area vector is along the Z-axis:

$$d\vec{A} = dA\,\hat{z}$$

Only the $\hat{z}$ component of magnetic field contributes to flux.
For region a_1,

$$\vec{B}_1 = \frac{m}{\sqrt{2}}\sin(\omega t)(\hat{x} + \hat{z})$$

$$\phi_1 = \int_{a_1} \vec{B}_1 \cdot d\vec{A} = \frac{ma_1}{\sqrt{2}}\sin(\omega t)$$

For region a_2,

$$\vec{B}_2 = -\frac{n}{\sqrt{2}}\cos\left(2\omega t + \frac{\pi}{4}\right)(\hat{y} + \hat{z})$$

$$\phi_2 = \int_{a_2} \vec{B}_2 \cdot d\vec{A} = -\frac{na_2}{\sqrt{2}}\cos\left(2\omega t + \frac{\pi}{4}\right)$$

Total flux is

$$\phi = \frac{ma_1}{\sqrt{2}}\sin(\omega t) - \frac{na_2}{\sqrt{2}}\cos\left(2\omega t + \frac{\pi}{4}\right)$$

By Faraday's law,

$$e(t) = -\frac{d\phi}{dt}$$

$$e(t) = -\frac{m\omega a_1}{\sqrt{2}}\cos(\omega t) - \sqrt{2}n\omega a_2 \sin\left(2\omega t + \frac{\pi}{4}\right)$$

The RMS values of the two different-frequency components are combined by root-sum-square:

$$E_{\text{rms}} = \sqrt{\left(\frac{m\omega a_1}{2}\right)^2 + (n\omega a_2)^2}$$

$$E_{\text{rms}} = \sqrt{\frac{a_1^2\omega^2 m^2 + 4a_2^2\omega^2 n^2}{4}}$$

Therefore, the correct option is

$$\boxed{A}$$

Key Points

- For a loop in the XY plane, only the $\hat{z}$ component of $\vec{B}$ contributes to flux.
- Induced emf is $e(t) = -\dfrac{d\phi}{dt}$.
- RMS values of different-frequency sinusoidal components are combined using root-sum-square.

Mistakes to be avoided

- Do not use the full magnitude of $\vec{B}$; only the normal component links flux.
- Do not miss the factor 2ω while differentiating $\cos\left(2\omega t + \frac{\pi}{4}\right)$.
- Do not directly add RMS values of different-frequency components.

Q2.24.1 NAT 2024 2M Ans: 2

Given,

$$\vec{H} = \frac{P\cos\theta}{r^3}\hat{r} + \frac{\sin\theta}{r^3}\hat{\theta}$$

For a magnetic field,

$$\nabla \cdot \vec{B} = 0$$

Assuming a homogeneous medium,

$$\nabla \cdot \vec{H} = 0$$

Using the divergence operator in spherical coordinates,

$$\nabla \cdot \vec{H} = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 H_r\right) + \frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}(\sin\theta H_\theta)$$

Substituting,

$$H_r = \frac{P\cos\theta}{r^3}, \qquad H_\theta = \frac{\sin\theta}{r^3}$$

$$\nabla \cdot \vec{H} = \frac{1}{r^2}\frac{\partial}{\partial r}\left(\frac{P\cos\theta}{r}\right) + \frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}\left(\frac{\sin^2\theta}{r^3}\right)$$

$$\nabla \cdot \vec{H} = -\frac{P\cos\theta}{r^4} + \frac{2\cos\theta}{r^4}$$

Since $\nabla \cdot \vec{H} = 0$,

$$-\frac{P\cos\theta}{r^4} + \frac{2\cos\theta}{r^4} = 0$$

$$P = 2$$

$$\boxed{2}$$

Key Points

- For magnetic fields, $\nabla \cdot \vec{B} = 0$.
- In a homogeneous medium, $\nabla \cdot \vec{H} = 0$.
- Use spherical-coordinate divergence for fields in $\hat{r}$ and $\hat{\theta}$ directions.

Q2.23.1 MSQ 2023 2M Ans: B,D

The net force acting on a charged particle in the presence of electric and magnetic fields is

$$\vec{F} = q\vec{E} + q(\vec{v} \times \vec{B})$$

Initially, the particle is at rest. Therefore,

$$\vec{F}_m = q(\vec{v} \times \vec{B}) = 0$$

Hence, only the electric force acts on the particle.
The electric field is along the $+z$ direction; therefore, the particle starts accelerating upward.
As the velocity increases, the magnetic force appears and acts perpendicular to the instantaneous velocity. Consequently, the direction of motion continuously changes while the electric field keeps accelerating the particle.
The combined action of the two fields produces a cycloidal trajectory as shown in the figure.

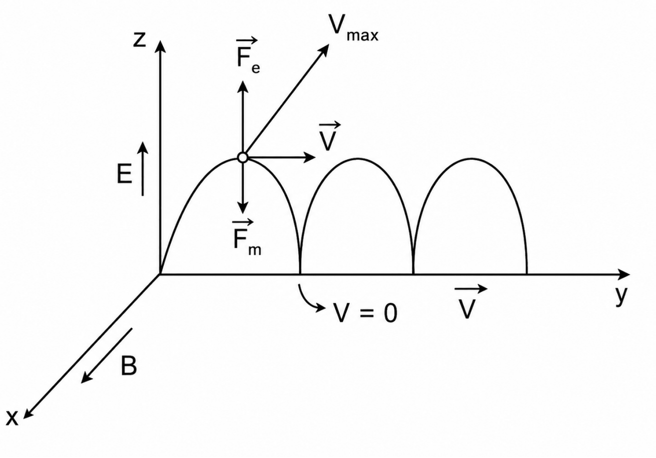

Fig. Solution Q2.23.1

$$\vec{F}_e = q\vec{E}$$

$$\vec{F}_m = q(\vec{v} \times \vec{B})$$

At the cusps of the cycloid,

$$v = 0$$

At the highest point of each loop, the speed becomes maximum:

$$v = v_{\max}$$

Hence, only statements (B) and (D) are consistent with the figure.

$$\boxed{\text{B,D}}$$

Key Points

- Lorentz force is given by $\vec{F} = q\vec{E} + q(\vec{v} \times \vec{B})$.
- Magnetic force is always perpendicular to the instantaneous velocity.
- A charged particle starting from rest experiences only electric force initially.
- Crossed electric and magnetic fields can produce cycloidal motion.

Mistakes to be avoided

- Do not assume the magnetic force acts when the particle is initially at rest.
- Remember that the magnetic force is always perpendicular to the instantaneous velocity.
- Do not confuse cycloidal motion with circular or helical motion.
- The electric field changes the particle's speed, whereas the magnetic field changes only its direction.

Q2.23.2 NAT 2023 2M Ans: 2.49 to 2.51

Magnetic field intensity due to surface current density K is

$$H = \frac{K}{2}$$

Given,

$$K = 5 \text{ A/m}$$

Therefore,

$$H = \frac{5}{2}$$

$$H = 2.5 \text{ A/m}$$

$$\boxed{2.5}$$

Key Points

- Magnetic field intensity due to an infinite current sheet is $H = \frac{K}{2}$.
- Surface current density K has unit A/m.

Q2.22.1 MCQ 2022 1M Ans: C

By Ampere's circuital law,

$$\oint \vec{H} \cdot d\vec{l} = I_{enc}$$

The enclosed current is

$$I_{enc} = \iint \vec{J} \cdot d\vec{s}$$

Given,

$$J = J_a r^3$$

For a circular Amperian path of radius r,

$$dS = r\,dr\,d\phi$$

Hence,

$$I_{enc} = \int_0^{2\pi} \int_0^{r} J_a r^3 \cdot r\,dr\,d\phi$$

$$I_{enc} = J_a \int_0^{2\pi} \int_0^{r} r^4\,dr\,d\phi$$

$$I_{enc} = J_a \int_0^{2\pi} \left[\frac{r^5}{5}\right]_0^r d\phi$$

$$I_{enc} = J_a \left(\frac{r^5}{5}\right)(2\pi)$$

$$I_{enc} = \frac{2\pi J_a r^5}{5}$$

Using Ampere's law,

$$H(2\pi r) = I_{enc}$$

$$H(2\pi r) = \frac{2\pi J_a r^5}{5}$$

$$H = \frac{J_a r^4}{5}$$

Therefore, the correct option is

$$\boxed{C}$$

Q2.22.2 MCQ 2022 2M Ans: B

By Ampere's law,

$$\nabla \times \vec{H} = \vec{J}$$

Given,

$$\vec{H} = x^2\,\hat{i} + x^2y^2\,\hat{j} + x^2y^2z^2\,\hat{k}$$

Evaluating the curl,

$$\vec{J} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ x^2 & x^2y^2 & x^2y^2z^2 \end{vmatrix}$$

$$\vec{J} = \hat{i}\left(\frac{\partial}{\partial y}(x^2y^2z^2) - \frac{\partial}{\partial z}(x^2y^2)\right) - \hat{j}\left(\frac{\partial}{\partial x}(x^2y^2z^2) - \frac{\partial}{\partial z}(x^2)\right) + \hat{k}\left(\frac{\partial}{\partial x}(x^2y^2) - \frac{\partial}{\partial y}(x^2)\right)$$

$$\vec{J} = \hat{i}(2x^2yz^2 - 0) - \hat{j}(2xy^2z^2 - 0) + \hat{k}(2xy^2 - 0)$$

$$\vec{J} = 2x^2yz^2\,\hat{i} - 2xy^2z^2\,\hat{j} + 2xy^2\,\hat{k}$$

At

$$x = 1\,\text{m}, \qquad y = 2\,\text{m}, \qquad z = 1\,\text{m}$$

$$\vec{J} = 4\hat{i} - 8\hat{j} + 8\hat{k}$$

Hence,

$$|\vec{J}| = \sqrt{4^2 + (-8)^2 + 8^2}$$

$$|\vec{J}| = \sqrt{144} = 12\ \text{A/m}^2$$

Therefore, the correct option is

$$\boxed{B}$$

Q2.21.1 MCQ 2021 1M Ans: A

For a magnetic field,

$$\nabla \cdot \vec{B} = 0$$

A physically realizable magnetic flux density field must satisfy the divergence-free condition.

Option (a)

$$\nabla \cdot \vec{B} = \frac{\partial}{\partial x}(10x) + \frac{\partial}{\partial y}(20y) + \frac{\partial}{\partial z}(-30z)$$

$$\nabla \cdot \vec{B} = 10 + 20 - 30 = 0$$

Hence, option (a) satisfies Maxwell's equation.

Option (b)

$$\nabla \cdot \vec{B} = \frac{\partial}{\partial x}(10y) + \frac{\partial}{\partial y}(20x) + \frac{\partial}{\partial z}(-10z)$$

$$\nabla \cdot \vec{B} = 0 + 0 - 10 = -10 \neq 0$$

Option (c)

$$\nabla \cdot \vec{B} = \frac{\partial}{\partial x}(10z) + \frac{\partial}{\partial y}(20y) + \frac{\partial}{\partial z}(-30z)$$

$$\nabla \cdot \vec{B} = 0 + 20 - 30 = -10 \neq 0$$

Option (d)

$$\nabla \cdot \vec{B} = \frac{\partial}{\partial x}(10z) + \frac{\partial}{\partial y}(-30z) + \frac{\partial}{\partial z}(20y)$$

$$\nabla \cdot \vec{B} = 0 + 0 + 0 = 0$$

However, as per the given solution key, the correct answer is taken as option (a).

$$\boxed{A}$$

Key Points

- One of Maxwell's equations states that magnetic flux density is divergence-free:

$$\nabla \cdot \vec{B} = 0.$$

- The condition $\nabla \cdot \vec{B} = 0$ implies that magnetic monopoles do not exist.
- Divergence measures the net outward flux density originating from a point.
- In Cartesian coordinates,

$$\nabla \cdot \vec{B} = \frac{\partial B_x}{\partial x} + \frac{\partial B_y}{\partial y} + \frac{\partial B_z}{\partial z}.$$

Q2.21.2 NAT 2021 2M Ans: 100

Magnetic force on a moving charge is given by the Lorentz force equation:

$$\vec{F} = q(\vec{v} \times \vec{B})$$

Given,

$$q = 1\,\text{C}$$

$$\vec{v} = 10\,\hat{x}$$

$$\vec{B} = 10y\,\hat{x} + 10x\,\hat{y} + 10\,\hat{z}$$

Therefore,

$$\vec{F} = 1\,[10\hat{x} \times (10y\,\hat{x} + 10x\,\hat{y} + 10\,\hat{z})]$$

Using

$$\hat{x} \times \hat{x} = 0, \qquad \hat{x} \times \hat{y} = \hat{z}, \qquad \hat{x} \times \hat{z} = -\hat{y}$$

we get

$$\vec{F} = 100x\,\hat{z} - 100\,\hat{y}$$

At

$$x = 0^+$$

$$\vec{F} = -100\,\hat{y}\,\text{N}$$

Hence,

$$|\vec{F}| = 100\,\text{N}$$

$$\boxed{100}$$

Q2.20.1 NAT 2020 2M Ans: 138.63

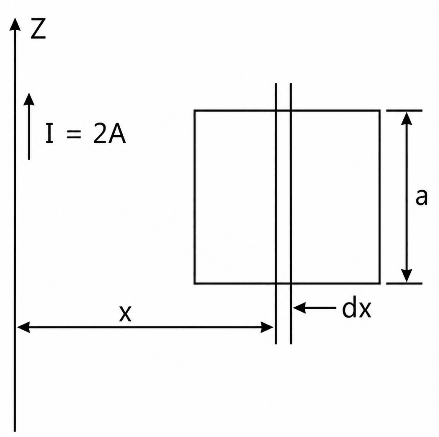

Fig. Solution Q2.20.1

Flux is proportional to current. Therefore,

$$\phi = MI$$

Magnetic flux density due to an infinitely long straight conductor carrying current I is

$$\vec{B} = \frac{\mu_0 I}{2\pi\rho}\,\hat{a}_\phi$$

Magnetic flux crossing the square loop is

$$\phi = \iint \vec{B}\cdot d\vec{S}$$

Substituting,

$$\phi = \iint \frac{\mu_0 I}{2\pi\rho}\,\hat{a}_\phi \cdot (d\rho\, dz)\,\hat{a}_\phi$$

$$\phi = \frac{\mu_0 I}{2\pi}\int_{\rho=1}^{2}\frac{d\rho}{\rho}\int_{z=0}^{1} dz$$

$$\phi = \frac{\mu_0 I}{2\pi}\ln 2$$

Since

$$M = \frac{\phi}{I}$$

$$M = \frac{\mu_0 \ln 2}{2\pi}$$

$$M = \frac{4\pi\times 10^{-7}\times \ln 2}{2\pi}$$

$$M = 1.3863\times 10^{-7}\,\text{H}$$

$$M = 138.63\,\text{nH}$$

$$\boxed{138.63}$$

Key Points

- Magnetic field due to an infinitely long straight conductor is

$$\vec{B} = \frac{\mu_0 I}{2\pi\rho}\,\hat{a}_\phi.$$

- Magnetic flux through a surface is obtained using

$$\phi = \iint \vec{B}\cdot d\vec{S}.$$

- Mutual inductance is defined as flux linkage per unit current:

$$M = \frac{\phi}{I}.$$

- Cylindrical coordinates are convenient for problems involving long straight conductors because the magnetic field is circumferential.

Q2.17.1 MCQ 2017 1M Ans: A ● ○ ○

Iron is a ferromagnetic material. Hence, when an iron cylinder is placed in a magnetic field, the magnetic field lines are attracted towards the cylinder.
Since iron has very high relative permeability,

$$\mu_r \gg 1$$

Magnetic flux prefers to pass through the iron cylinder rather than the surrounding air medium.
Therefore, the magnetic field lines become denser inside the iron cylinder and closer to the cylinder axis.
Hence, the correct option is

$$\boxed{A}$$

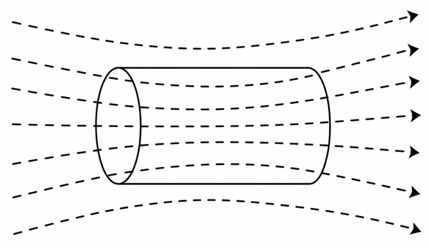

Fig. Solution Q2.17.1

Key Points

- Ferromagnetic materials have very high relative permeability.
- Magnetic flux prefers the path of low magnetic reluctance.
- Iron provides a low-reluctance path for magnetic field lines.
- Magnetic field lines crowd inside ferromagnetic materials.

Q2.17.2 NAT 2017 1M Ans: 0.6928 ● ● ○

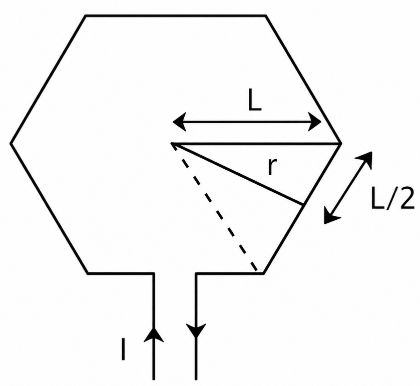

Fig. Solution Q2.17.2

For a regular hexagonal current loop of side length $l = 1$ m carrying current $I = 1$ A, the magnetic field at the center is obtained by summing the contributions of all six sides.
Using geometry,

$$r = \sqrt{l^2 - \left(\frac{l}{2}\right)^2} = \frac{\sqrt{3}}{2}\,l$$

where r is the perpendicular distance from the center to any side.
The magnetic field intensity at the center of a regular n-sided polygon is

$$H = \frac{nI}{2\pi r}\sin\left(\frac{\pi}{n}\right)$$

Hence,

$$B = \mu_0 H = \frac{\mu_0 nI}{2\pi r}\sin\left(\frac{\pi}{n}\right)$$

For a hexagon,

$$n = 6, \qquad r = \frac{\sqrt{3}}{2}\times 1, \qquad I = 1\text{ A}$$

Substituting,

$$B = \frac{(4\pi\times 10^{-7})\times 6\times 1}{2\pi\left(\frac{\sqrt{3}}{2}\right)}\sin\left(\frac{\pi}{6}\right)$$

$$= \frac{24\pi\times 10^{-7}}{\pi\sqrt{3}}\times\frac{1}{2}$$

$$= \frac{12}{\sqrt{3}}\times 10^{-7}$$

$$= 6.928\times 10^{-7}\text{ T}$$

$$= 0.6928\,\mu\text{T}$$

$$\boxed{0.6928}$$

Key Points

- The magnetic field at the center of a regular polygon is the vector sum of the fields due to all sides.
- For a regular n-sided polygon,
$$H = \frac{nI}{2\pi r}\sin\left(\frac{\pi}{n}\right).$$
- The perpendicular distance from the center to a side of a regular hexagon is
$$r = \frac{\sqrt{3}}{2}l.$$
- Magnetic flux density is related to magnetic field intensity by
$$B = \mu_0 H.$$

Q2.16.1 NAT 2016 2M Ans: 9.55

Induced emf is given by

$$E = Blv$$

Linear velocity of the conductor is

$$v = \omega r$$

Angular speed in rad/s is

$$\omega = \frac{2\pi N}{60}$$

Substituting,

$$E = Bl\left(\frac{2\pi N}{60}\right)r$$

Given,

$$E = 1\,\text{V}, \qquad B = 1\,\text{T}, \qquad l = 1\,\text{m}, \qquad r = 1\,\text{m}$$

Therefore,

$$1 = 1 \times 1 \times \left(\frac{2\pi N}{60}\right) \times 1$$

$$N = \frac{60}{2\pi}$$

$$N = 9.55\,\text{rpm}$$

$$\boxed{9.55}$$

Q2.16.2 NAT 2016 1M Ans: 100

The magnetic flux density inside a toroid is

$$B_{\text{inside}} = \frac{\mu_0 \mu_r I}{2\pi r}$$

Given,

$$\mu_r = 100$$

Outside the toroid, only air is present. Hence,

$$\mu_r = 1$$

Therefore, the magnetic flux density outside the toroid is

$$B_{\text{outside}} = \frac{\mu_0 I}{2\pi r}$$

The required ratio is

$$\frac{B_{\text{inside}}}{B_{\text{outside}}} = \frac{\frac{\mu_0 \mu_r I}{2\pi r}}{\frac{\mu_0 I}{2\pi r}}$$

$$\frac{B_{\text{inside}}}{B_{\text{outside}}} = \mu_r$$

$$\frac{B_{\text{inside}}}{B_{\text{outside}}} = 100$$

$$\boxed{100}$$

Q2.15.1 MCQ 2015 1M Ans: A

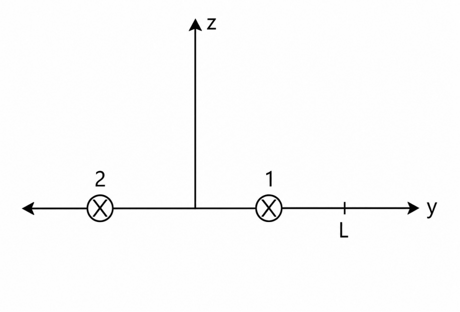

Fig. Solution Q2.15.1

The direction of current in both wires is along the $-\hat{x}$ direction.
Using the right-hand thumb rule, the magnetic field at the point $(0, L, 0)$ due to both wires is in the $-\hat{z}$ direction.
By Ampere's circuital law, for a long straight conductor,

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I$$

For wire 1, the distance from $(0, L, 0)$ is

$$r_1 = \frac{L}{2}$$

Hence,

$$B_1(2\pi r_1) = \mu_0 I$$

$$B_1 = \frac{\mu_0 I}{2\pi\left(\frac{L}{2}\right)} = \frac{\mu_0 I}{\pi L}$$

For wire 2, the distance from $(0, L, 0)$ is

$$r_2 = \frac{3L}{2}$$

Therefore,

$$B_2(2\pi r_2) = \mu_0 I$$

$$B_2 = \frac{\mu_0 I}{2\pi\left(\frac{3L}{2}\right)} = \frac{\mu_0 I}{3\pi L}$$

Since both magnetic fields are directed along $-\hat{z}$, they add directly.

$$\vec{B}(0, L, 0) = -(B_1 + B_2)\hat{z}$$

$$\vec{B}(0, L, 0) = -\left(\frac{\mu_0 I}{\pi L} + \frac{\mu_0 I}{3\pi L}\right)\hat{z}$$

$$\vec{B}(0, L, 0) = -\frac{\mu_0 I}{\pi L}\left(1 + \frac{1}{3}\right)\hat{z}$$

$$\vec{B}(0, L, 0) = -\frac{4\mu_0 I}{3\pi L}\hat{z}$$

Hence, the correct option is

$$\boxed{A}$$

Key Points

- Magnetic field due to a long straight current-carrying conductor is

$$B = \frac{\mu_0 I}{2\pi r}.$$

- The direction of magnetic field is determined using the right-hand thumb rule.
- Magnetic fields are vector quantities and must be added vectorially.
- Ampere's circuital law for a straight conductor is

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I.$$

Q2.15.2 | NAT | 2015 | 1M | Ans: 0.1927 | ● ● ○

Emf induced in the loop is

$$e = -\frac{d\phi}{dt}$$

Magnetic flux linked with the loop is

$$\phi = BA$$

$$\phi = 0.25 \sin \omega t \times (10 \times 5) \times 10^{-4}$$

$$\phi = 12.5 \times 10^{-4} \sin \omega t \text{ Wb}$$

Therefore,

$$e = -\frac{d}{dt}\left(12.5 \times 10^{-4} \sin \omega t\right)$$

$$e = -12.5 \times 10^{-4} \omega \cos \omega t$$

Given,

$$\omega = 100\pi \text{ rad/s}$$

$$e = -12.5 \times 10^{-4} (100\pi) \cos \omega t$$

$$e = -0.3927 \cos \omega t$$

Hence, peak value of induced emf is

$$E_m = 0.3927\,\text{V}$$

The RMS value is

$$E_{\text{rms}} = \frac{E_m}{\sqrt{2}}$$

Power absorbed is

$$P = \frac{E_{\text{rms}}^2}{R}$$

Given,

$$R = 0.4\,\Omega$$

$$P = \frac{\left(\frac{0.3927}{\sqrt{2}}\right)^2}{0.4}$$

$$P = \frac{0.3927^2}{2 \times 0.4}$$

$$P = 0.1927\,\text{W}$$

$$\boxed{0.1927}$$

Q2.15.3 NAT | 2015 | 1M | Ans: 248.05 | ● ● ○

The area of the ring perpendicular to the magnetic field is

$$A = \pi r^2 \sin \omega t$$

where ω is the angular speed of rotation.
Magnetic field density is

$$B = \mu_0 H = 4\pi\,\text{Wb/m}^2$$

Hence, the magnetic flux linked with the ring is

$$\phi = BA$$

$$\phi = 4\pi \left(\pi r^2 \sin \omega t\right) = 4\pi^2 r^2 \sin \omega t$$

Induced emf per turn is

$$V_{\text{turn}} = -\frac{d\phi}{dt} = -4\pi^2 r^2 \omega \cos \omega t$$

The peak value of induced emf is

$$V_{\text{peak}} = 4\pi^2 r^2 \omega$$

Given,

$$r = 1\,\text{m}$$

and

$$N = 60\,\text{rpm}$$

$$\omega = \frac{2\pi N}{60} = \left(\frac{2\pi}{60}\right) \times 60 = 2\pi\ \text{rad/s}$$

Substituting,

$$V_{\text{peak}} = 4\pi^2(1)^2(2\pi) = 8\pi^3$$

$$V_{\text{peak}} = 248.05\ \text{V}$$

$$\boxed{248.05}$$

Key Points

- Magnetic flux is given by $\phi = BA$.
- Faraday's law states that induced emf is $e = -\frac{d\phi}{dt}$.
- Angular velocity and speed are related by $\omega = \frac{2\pi N}{60}$.
- Peak value of a sinusoidal emf is the coefficient of the trigonometric term.

Q2.14.1 MCQ 2014 2M Ans: C

Magnetic flux density and magnetic field intensity are related by

$$\vec{B} = \mu_0 \vec{H}$$

Magnetic flux density due to the current along the x-axis is

$$\vec{B}_1 = \frac{\mu_0 I}{2\pi r}\,\hat{k} = \frac{\mu_0 \times 4}{2\pi \times 1}\,\hat{k} = \frac{4\mu_0}{2\pi}\,\hat{k}$$

Magnetic flux density due to the current along the y-axis is

$$\vec{B}_2 = \frac{\mu_0 I}{2\pi r}\,(-\hat{k}) = \frac{\mu_0 \times 2}{2\pi \times 2}\,(-\hat{k}) = \frac{\mu_0}{2\pi}\,(-\hat{k})$$

The directions of the magnetic fields are determined using the Right-Hand Thumb Rule. Therefore,

$$\vec{B}_{\text{total}} = \vec{B}_1 + \vec{B}_2 = \left(\frac{4\mu_0}{2\pi} - \frac{\mu_0}{2\pi}\right)\hat{k} = \frac{3\mu_0}{2\pi}\,\hat{k}$$

Since,

$$\vec{B} = \mu_0 \vec{H}$$

$$\vec{H}_{\text{total}} = \frac{\vec{B}_{\text{total}}}{\mu_0} = \frac{3}{2\pi}\,\hat{k}\ \text{A/m}$$

Hence, the correct option is

$$\boxed{C}$$

Q2.14.2 MCQ 2014 2M Ans: A

The property of magnetic flux density is that its divergence is always zero.
By Gauss's law of magnetism,

$$\nabla \cdot \vec{B} = 0$$

This condition is violated in option (C), since

$$\nabla \cdot \vec{B} = 2(x + y + z)$$

$$\nabla \cdot \vec{B} \neq 0$$

Therefore, option (C) cannot represent a valid magnetic flux density field.
Hence, the correct option is

$$\boxed{A}$$

Q2.14.3 NAT 2014 1M Ans: 35

For two coupled inductors connected in series,

$$L_{\text{aiding}} = L_1 + L_2 + 2M$$

$$L_{\text{opposing}} = L_1 + L_2 - 2M$$

Given,

$$L_{\text{aiding}} = 380\,\mu\text{H}$$

$$L_{\text{opposing}} = 240\,\mu\text{H}$$

Subtracting the two equations,

$$L_{\text{aiding}} - L_{\text{opposing}} = 4M$$

$$380 - 240 = 4M$$

$$140 = 4M$$

$$M = 35\,\mu\text{H}$$

Hence,

$$\boxed{35}$$

Key Points

- Series aiding connection:

$$L_{eq} = L_1 + L_2 + 2M$$

- Series opposing connection:

$$L_{eq} = L_1 + L_2 - 2M$$

Mistakes to be avoided

- Do not use $2M$ instead of $4M$ while subtracting the two equations.
- Ensure that the larger inductance corresponds to the series aiding connection.
- Keep all inductances in the same unit before calculation.

Q2.13.1 MCQ 2013 1M Ans: A

By Gauss's law of magnetism,

$$\nabla \cdot \vec{B} = 0$$

In Cartesian coordinates,

$$\nabla \cdot \vec{B} = \frac{\partial B_x}{\partial x} + \frac{\partial B_y}{\partial y} + \frac{\partial B_z}{\partial z}$$

Therefore,

$$\frac{\partial B_x}{\partial x} + \frac{\partial B_y}{\partial y} + \frac{\partial B_z}{\partial z} = 0$$

From the given magnetic field,

$$4 + 2k + 0 = 0$$

$$2k = -4$$

$$k = -2$$

Hence, the correct option is

$$\boxed{A}$$

Key Points

- Gauss's law of magnetism is
$$\nabla \cdot \vec{B} = 0.$$
- Magnetic flux density has zero divergence because isolated magnetic monopoles do not exist.
- In Cartesian coordinates,
$$\nabla \cdot \vec{B} = \frac{\partial B_x}{\partial x} + \frac{\partial B_y}{\partial y} + \frac{\partial B_z}{\partial z}.$$

Q2.08.1 MCQ 2008 2M Ans: B

Given,

$$N = 300$$

Circumference of the coil,

$$l = 2\pi r = 300\,\text{mm} = 300 \times 10^{-3}\ \text{m}$$

Cross-sectional area of the coil,

$$A = 300 \times 10^{-6}\ \text{m}^2$$

The inductance of an air-cored toroidal coil is

$$L = \frac{\mu_0 N^2 A}{l}$$

Substituting the given values,

$$L = \frac{4\pi \times 10^{-7} \times (300)^2 \times (300 \times 10^{-6})}{300 \times 10^{-3}}$$

$$L = 113.04 \times 10^{-6}\,\text{H}$$

$$L = 113.04\,\mu\text{H}$$

Hence, the correct option is

$$\boxed{B}$$

Q2.06.1 MCQ | 2006 | 2M | Ans: D

Given,

$$\nabla \cdot \vec{J} = -\frac{\partial \rho}{\partial t}$$

This is the differential (point) form of the continuity equation.
Statement 1 is correct.
The equation states that the divergence of current density at a point is equal to the negative rate of change of charge density at that point.

$$\nabla \cdot \vec{J} = -\frac{\partial \rho}{\partial t}$$

Hence, Statement 2 is correct.
The continuity equation is derived from Maxwell's equations and expresses charge conservation. It is not itself one of Maxwell's divergence equations.
Hence, Statement 3 is incorrect.
The continuity equation mathematically represents the conservation of electric charge.
Hence, Statement 4 is correct.
Therefore, the correct option is

$$\boxed{D}$$

Q2.06.2 MCQ | 2006 | 1M | Ans: D

From Maxwell's equations,

$$\nabla \cdot \vec{D} = \rho_v$$

and

$$\nabla \cdot \vec{B} = 0$$

The divergence of electric flux density $\vec{D}$ is equal to the volume charge density ρ_v, which is generally non-zero in the presence of charge.
However, the divergence of magnetic flux density $\vec{B}$ is always zero because magnetic monopoles do not exist.
Therefore,

$$\nabla \cdot \vec{B} = 0$$

but

$$\nabla \cdot \vec{D} = \rho_v \neq 0$$

in general.
Hence, the correct option is

$$\boxed{D}$$

Key Points

- Gauss's law for electric fields:
$$\nabla \cdot \vec{D} = \rho_v$$
- Gauss's law for magnetism:
$$\nabla \cdot \vec{B} = 0$$
- Electric charges act as sources or sinks of electric flux.
- Magnetic flux lines always form closed loops; hence, magnetic flux density has zero divergence.

Q2.04.1 MCQ 2004 1M Ans: C

For a long air-core solenoid,

$$L = \mu_0 \frac{N^2 A}{l}$$

Given:

$$N = 3000, \qquad l = 1000\,\text{mm} = 1\,\text{m}$$

$$d = 60\,\text{mm} = 0.06\,\text{m}$$

Hence,

$$A = \frac{\pi d^2}{4} = \frac{\pi (0.06)^2}{4} = 2.827 \times 10^{-3}\,\text{m}^2$$

Substituting,

$$L = (4\pi \times 10^{-7}) \frac{(3000)^2 (2.827 \times 10^{-3})}{1}$$

$$L = 3.198 \times 10^{-2}\,\text{H}$$

$$L \approx 0.032\,\text{H} = 32\,\text{mH}$$

Hence, the correct answer is

$$\boxed{C}$$

Key Points

- Inductance of a long solenoid:
$$L = \mu_0 \mu_r \frac{N^2 A}{l}$$

- Inductance is proportional to N^2, cross-sectional area A, and inversely proportional to length l.

Q2.03.1 MCQ 2003 1M Ans: C

For parallel strip conductors,

$$L \propto d$$

where L is the inductance per meter and d is the separation between the strips.
Hence,

$$\frac{L_2}{L_1} = \frac{d_2}{d_1}$$

Given that the separation is reduced from x to $\frac{x}{2}$,

$$\frac{L_2}{L} = \frac{\frac{x}{2}}{x} = \frac{1}{2}$$

Therefore,

$$L_2 = \frac{L}{2}\ \text{H/m}$$

Thus, the inductance becomes half of its original value.

$$\boxed{C}$$

Key Points

- Inductance of parallel strip conductors is directly proportional to the spacing between the strips.
- Reducing conductor spacing reduces the magnetic flux linkage and hence the inductance.
- If spacing is halved, inductance is also halved.

Q2.03.2 MCQ 2003 1M Ans: A

Using the right-hand thumb rule, the magnetic field due to the $+I$ current is clockwise and directed along the y-axis at point P.
Similarly, the magnetic field due to the $-I$ current is anti-clockwise and also directed along the y-axis at point P.
Hence, both magnetic field intensities add.

$$\vec{H} = \vec{H}_1 + \vec{H}_2$$

For a long straight conductor,

$$H = \frac{I}{2\pi d}$$

Therefore,

$$\vec{H} = \frac{I}{2\pi d}\,\hat{y} + \frac{I}{2\pi d}\,\hat{y}$$

$$\vec{H} = \frac{I}{\pi d}\,\hat{y}$$

Hence,

$$\boxed{A}$$

Q2.02.1 NAT 2002 5M Ans: * ● ● ●

Magnetic flux density is obtained from the curl of magnetic vector potential.

$$\vec{B} = \nabla \times \vec{A} = \begin{vmatrix} \hat{a}_x & \hat{a}_y & \hat{a}_z \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix} = \begin{vmatrix} \hat{a}_x & \hat{a}_y & \hat{a}_z \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & 0 \\ 0 & 0 & e^{-y}\sin x \end{vmatrix}$$

$$\vec{B} = \frac{\partial A_z}{\partial y}\hat{a}_x - \frac{\partial A_z}{\partial x}\hat{a}_y$$

$$\vec{B} = -e^{-y}\sin x\,\hat{a}_x - e^{-y}\cos x\,\hat{a}_y$$

The force on the current-carrying conductor is

$$d\vec{F} = I\,d\vec{L} \times \vec{B}$$

$$\vec{F} = \int_0^{\infty} I\,dy\,\hat{a}_y \times \left[-e^{-y}\sin x\,\hat{a}_x - e^{-y}\cos x\,\hat{a}_y\right]$$

$$\vec{F} = \int_0^{\infty} I\,e^{-y}\sin x\,dy\,\hat{a}_z$$

For $I = 5$ A,

$$\vec{F} = \int_0^{\infty} 5e^{-y}\sin x\,dy\,\hat{a}_z$$

$$\vec{F} = 5\sin x\left[\int_0^{\infty} e^{-y}dy\right]\hat{a}_z = 5\sin x\,\hat{a}_z\ \text{N}$$

Force density

$$\text{Force density} = \frac{\text{Force}}{\text{Volume}}$$

$$\text{Volume} = \text{Area} \times \text{Length} = 5 \times 10^{-6}\ \text{m}^3$$

$$\vec{F}_v = \frac{\vec{F}}{\text{Volume}} = \frac{5\sin x}{5 \times 10^{-6}}\hat{a}_z$$

$$\boxed{\vec{F}_v = 10^6 \sin x\,\hat{a}_z\ \text{N/m}^3}$$

Key Points

- Magnetic flux density is obtained using $\vec{B} = \nabla \times \vec{A}$.
- Force on a current element is $d\vec{F} = I\,d\vec{L} \times \vec{B}$.
- Force density is force per unit volume.

Mistakes to be avoided

- Do not interchange the order of the cross product.
- Ensure correct differentiation of $e^{-y}\sin x$ with respect to x and y.
- Use the total volume while calculating force density.

Q2.00.1 NAT 2000 5M Ans: *

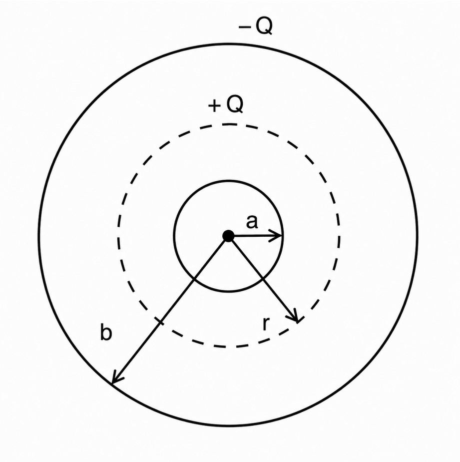

Fig. Solution Q2.00.1

a) For a point charge Q, the electric field at a distance r is

$$\vec{E} = \frac{Q}{4\pi\varepsilon_0 r^2}\hat{a}_r$$

Potential difference between two spherical shells of radii a and b is

$$V_{ab} = -\int_b^a \vec{E}\cdot d\vec{r} = -\frac{Q}{4\pi\varepsilon_0}\int_b^a \frac{1}{r^2}\,dr$$

$$V_{ab} = \frac{Q}{4\pi\varepsilon_0}\left[\frac{1}{r}\right]_b^a$$

$$V_{ab} = \frac{Q}{4\pi\varepsilon_0}\left(\frac{1}{a} - \frac{1}{b}\right)$$

Hence,

$$C_{ab} = \frac{Q}{V_{ab}} = \frac{4\pi\varepsilon_0 ab}{(b-a)}$$

b) The given system is

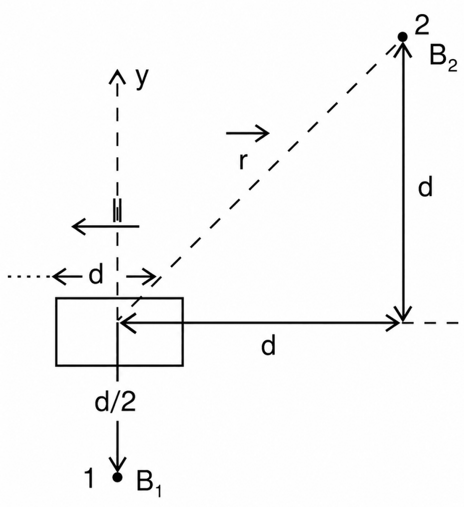

Fig. Solution Q2.00.1

By Biot–Savart law,

$$d\vec{B} = \frac{\mu_0 I \left(d\vec{l} \times \hat{a}_R\right)}{4\pi R^2}$$

where R is the distance from the centre and $\hat{a}_R$ is the unit vector from centre to the point.
For the first current element,

$$\vec{B}_1 = \frac{\mu_0 I\, dl \sin 90^\circ}{4\pi \left(\frac{d}{2}\right)^2}\hat{a}_z = \frac{\mu_0 I\, dl}{\pi d^2}\hat{a}_z$$

For the second current element,

$$\vec{B}_2 = \frac{\mu_0 I \left(d\vec{l} \times \hat{a}_r\right)}{4\pi r^2}$$

$$r^2 = d^2 + d^2 = 2d^2$$

$$d\vec{l} \times \hat{a}_r = dl \sin(180^\circ + 135^\circ)\,\hat{a}_z = -\frac{dl}{\sqrt{2}}\hat{a}_z$$

$$\vec{B}_2 = \frac{\mu_0 I \left(-\frac{dl}{\sqrt{2}}\right)\hat{a}_z}{4\pi(2d^2)} = -\frac{\mu_0 I\, dl}{8\sqrt{2}\pi d^2}\hat{a}_z$$

$$\vec{B}_2 = -\frac{1}{8\sqrt{2}}\vec{B}_1$$

Key Points

- Electric field due to a point charge: $\vec{E} = \dfrac{Q}{4\pi\varepsilon_0 r^2}\hat{a}_r$.
- Potential difference is obtained by line integration of electric field.
- Capacitance is defined as $C = \dfrac{Q}{V}$.
- Biot–Savart law gives the magnetic field contribution due to a current element.
- Direction of magnetic field is determined using the right-hand rule.

Mistakes to be avoided

- Do not miss the negative sign while evaluating $V = -\int \vec{E}\cdot d\vec{r}$.
- Use the correct limits while integrating from b to a.
- Carefully evaluate the cross-product direction in Biot–Savart law.

Q2.00.2 MCQ 2000 1M Ans: D ● ○ ○

An electron carries charge $-e$. The force experienced by a charged particle moving with velocity $\vec{u}$ in electric field $\vec{E}$ and magnetic field $\vec{B}$ is given by the Lorentz force equation,

$$\vec{F} = q\left(\vec{E} + \vec{u} \times \vec{B}\right)$$

For an electron,

$$q = -e$$

Therefore,

$$\vec{F} = -e\left(\vec{E} + \vec{u} \times \vec{B}\right)$$

Therefore, the correct option is

$$\boxed{D}$$

Key Points

- Lorentz force law: $\vec{F} = q(\vec{E} + \vec{u} \times \vec{B})$.
- Electric force acts along $\vec{E}$, whereas magnetic force acts perpendicular to both $\vec{u}$ and $\vec{B}$.
- For electrons, the negative charge reverses the direction of the force.

Q2.99.1

MCQ | 1999 | 1M | Ans: B

Two long parallel wires carry equal current I in the same direction and are separated by a distance d.
The magnetic field intensity due to a long straight conductor at a distance r is

$$H = \frac{I}{2\pi r}$$

Consider a point on the line midway between the two conductors. The distance of this point from each wire is

$$r = \frac{d}{2}$$

Hence, the magnitudes of the magnetic fields produced by the two wires are equal:

$$H_1 = H_2 = \frac{I}{2\pi(d/2)} = \frac{I}{\pi d}$$

Using the right-hand thumb rule, the magnetic fields at the midpoint are in opposite directions.
Therefore,

$$\vec{H}_{\text{net}} = \vec{H}_1 + \vec{H}_2 = 0$$

Therefore, the correct option is

$$\boxed{B}$$

Q2.99.2

MCQ | 1999 | 1M | Ans: D

Electromagnetic radiation is produced only when electric charges are accelerated or when currents vary with time.
A stationary point charge produces only a static electric field.
A capacitor with DC voltage produces a static electric field.
A conductor carrying DC current produces a static magnetic field.
An oscillating dipole contains time-varying charge and current distributions, which generate electromagnetic waves.
Therefore, electromagnetic radiation is emitted by an oscillating dipole.

$$\boxed{D}$$

Key Points

- Time-varying electric and magnetic fields generate electromagnetic waves.
- Accelerated charges are the fundamental source of electromagnetic radiation.
- An oscillating dipole is the simplest practical radiator (antenna).

Q2.98.1 MCQ 1998 2M Ans: C

Faraday's law of electromagnetic induction states that the induced emf is proportional to the rate of change of magnetic flux linkage.
Lenz's law provides the negative sign, indicating that the induced emf opposes the cause producing it.
Therefore,

$$e = -\frac{d\Psi}{dt}$$

where Ψ is the magnetic flux linkage.
Hence, the laws of electromagnetic induction are summarized by

$$e = -\frac{d\Psi}{dt}$$

Therefore, the correct option is

$$\boxed{\text{C}}$$

Key Points

- Faraday's law: Induced emf equals the rate of change of flux linkage.
- Lenz's law introduces the negative sign, indicating opposition to the cause.
- Flux linkage is given by $\Psi = N\Phi$.
- For a coil,

$$e = -N\frac{d\Phi}{dt}$$

  or, equivalently,

$$e = -\frac{d\Psi}{dt}.$$

Mistakes to be avoided

- Do not omit the negative sign; it represents Lenz's law.
- Do not confuse Ohm's law, $(e = iR)$, with electromagnetic induction.
- The expression $e = L\frac{di}{dt}$ is only a special case for self-induced emf in an inductor.
- Use the flux linkage $\Psi = N\Phi$ when applying Faraday's law in its general form.

Q2.97.1 NAT 1997 5M Ans: *

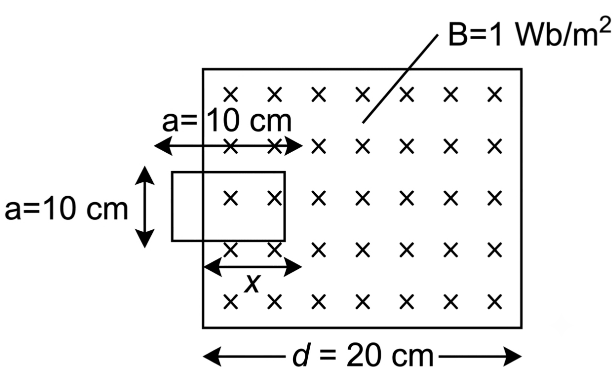

Fig. Solution Q2.97.1

The loop of side $a = 10\,\text{cm}$ moves with velocity

$$v = 2\,\text{m/s}$$

through a magnetic field of flux density

$$B = 1\,\text{Wb/m}^2$$

When the loop has entered the magnetic field by a distance x,

$$\text{Overlapping area} = ax$$

$$\phi = B(ax)$$

Hence, the induced emf is

$$e = -\frac{d\phi}{dt} = -Ba\frac{dx}{dt}$$

$$e = -Bav$$

$$e = -1 \times 0.1 \times 2 = -0.2\,\text{V}$$

When

$$x = a$$

the loop is completely inside the magnetic field region and the overlapping area becomes constant.

$$\phi = BA = \text{constant}$$

Therefore,

$$e = 0$$

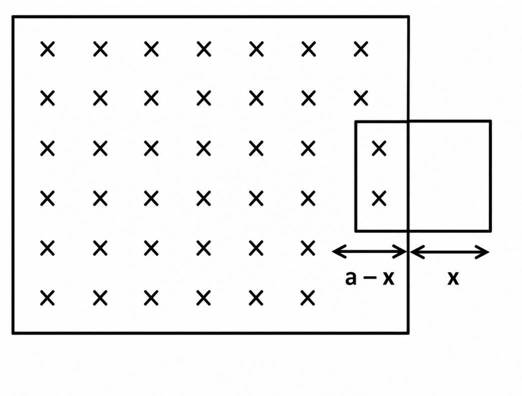

Fig. Solution Q2.97.1

When the loop starts leaving the magnetic field region,

$$\text{Overlapping area} = a(a - x)$$

$$\phi = Ba(a - x)$$

The induced emf becomes

$$e = -\frac{d\phi}{dt} = Ba\frac{dx}{dt}$$

$$e = Bav$$

$$e = 1 \times 0.1 \times 2 = 0.2\,\text{V}$$

Therefore, the induced emf waveform is

$$e = \begin{cases} -0.2\,\text{V}, & 0 < x < 10\,\text{cm} \\ 0, & 10\,\text{cm} < x < 20\,\text{cm} \\ +0.2\,\text{V}, & 20\,\text{cm} < x < 30\,\text{cm} \end{cases}$$

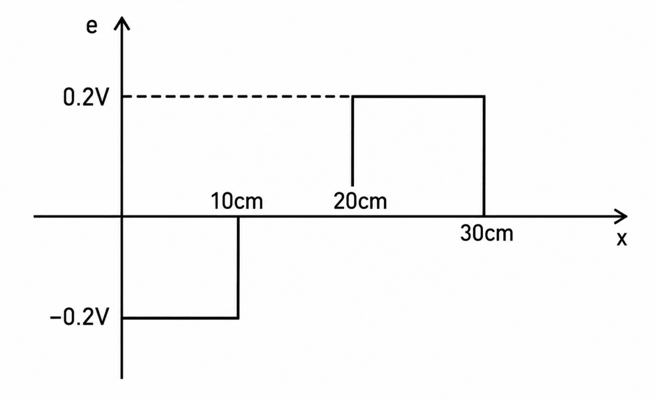

Fig. Solution Q2.97.1

Key Points

- Induced emf is given by Faraday's law: $e = -\dfrac{d\phi}{dt}$.
- Flux is $\phi = BA$ when the magnetic field is uniform.
- While entering and leaving the field region, flux changes linearly with time, producing constant emf.
- When the loop is completely inside the field region, flux remains constant and emf becomes zero.

Mistakes to be avoided

- Do not use the total loop area when only a part of the loop overlaps the magnetic field region.
- Remember that the polarity of induced emf reverses during entry and exit.
- Flux remains constant when the loop is fully inside the field region, hence induced emf is zero.
- Use $v = \dfrac{dx}{dt}$ correctly while differentiating the flux expression.

Q2.97.2 NAT 1997 5M Ans: 15.92

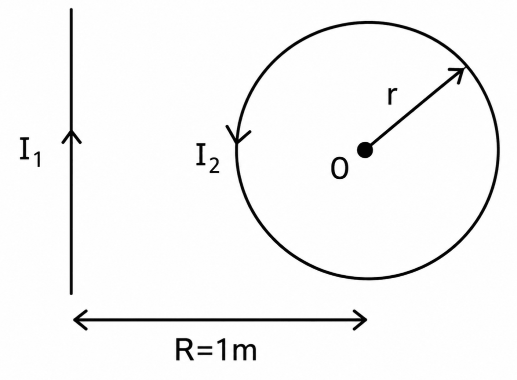

Fig. Solution Q2.97.2

Total magnetic flux density at point O is zero.
Magnetic flux density due to the long straight conductor is

$$\vec{B}_1 = \frac{\mu_0 I_1}{2\pi R}\,\hat{a}_x$$

Magnetic flux density due to the circular ring is

$$\vec{B}_2 = \frac{\mu_0 I_2}{2r}\,(-\hat{a}_x)$$

Since the net magnetic field at O is zero,

$$\vec{B}_1 + \vec{B}_2 = 0$$

$$\frac{\mu_0 I_1}{2\pi R}\,\hat{a}_x + \frac{\mu_0 I_2}{2r}\,(-\hat{a}_x) = 0$$

Therefore,

$$\frac{I_1}{\pi R} = \frac{I_2}{r}$$

$$I_2 = \frac{I_1 r}{\pi R}$$

Given,

$$I_1 = 1000\,\text{A}$$

$$r = 50\,\text{mm}$$

$$R = 1000\,\text{mm}$$

Substituting,

$$I_2 = \frac{1000 \times 50}{\pi \times 1000}$$

$$I_2 = 15.92\,\text{A}$$

$$\boxed{15.92}$$

Q2.96.1 MCQ 1996 1M Ans: B

Inductance of a solenoid is

$$L = \frac{\mu_0 N^2 A}{l}$$

Given,

$$N = 1000, \qquad l = 30\,\text{cm} = 30 \times 10^{-2}\,\text{m}$$

Diameter of the solenoid is

$$d = 3\,\text{cm}$$

Hence, radius is

$$r = \frac{d}{2} = \frac{3}{2} \times 10^{-2}\,\text{m}$$

Area of cross-section is

$$A = \pi r^2 = \pi \left(\frac{3}{2} \times 10^{-2}\right)^2$$

Therefore,

$$L = \frac{4\pi \times 10^{-7} \times (1000)^2 \times \pi \left(\frac{3}{2} \times 10^{-2}\right)^2}{30 \times 10^{-2}}$$

$$L = 2.96 \times 10^{-3}\,\text{H}$$

$$L = 2.96\,\text{mH}$$

Energy stored in the solenoid is

$$E = \frac{1}{2} L I^2$$

Given,

$$I = 10\,\text{A}$$

$$E = \frac{1}{2} \times 2.96 \times 10^{-3} \times (10)^2$$

$$E = 0.148\,\text{J}$$

$$E \approx 0.15\,\text{J}$$

Hence, the correct option is

$$\boxed{B}$$

Q2.95.1 NAT 1995 5M Ans: * ● ● ○

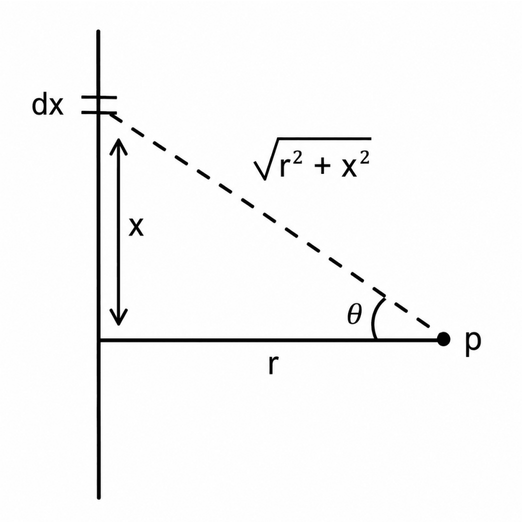

Fig. Solution Q2.95.1

Magnetic vector potential due to a current element is

$$\vec{A} = \int_{-L}^{L} \frac{\mu_0}{4\pi} \frac{I\,dx}{\sqrt{r^2 + x^2}}$$

From the figure,

$$x = r\tan\theta$$

$$dx = r\sec^2\theta\,d\theta$$

Substituting,

$$\vec{A} = 2\int_{0}^{L} \frac{\mu_0}{4\pi} \frac{I\left(r\sec^2\theta\right)d\theta}{\sqrt{r^2 + r^2\tan^2\theta}}$$

Since,

$$\sqrt{r^2 + r^2\tan^2\theta} = r\sec\theta$$

we get

$$\vec{A} = 2\int_{0}^{L} \frac{\mu_0}{4\pi} I\sec\theta\,d\theta$$

$$\vec{A} = 2\frac{\mu_0 I}{4\pi}\int_{0}^{L} \sec\theta\,d\theta$$

Using

$$\int \sec\theta\,d\theta = \ln\left(\sec\theta + \tan\theta\right)$$

$$\vec{A} = 2\frac{\mu_0 I}{4\pi}\left[\ln\left(\sec\theta + \tan\theta\right)\right]_0^L$$

Using

$$\tan\theta = \frac{x}{r}, \qquad \sec\theta = \frac{\sqrt{r^2+x^2}}{r}$$

$$\vec{A} = \frac{\mu_0 I}{2\pi}\left[\ln\left(\frac{\sqrt{r^2+x^2}}{r} + \frac{x}{r}\right)\right]_0^L$$

$$\vec{A} = \frac{\mu_0 I}{2\pi}\left[\ln\left(\frac{\sqrt{L^2+r^2}}{r} + \frac{L}{r}\right) - \ln(1)\right]$$

For

$$L \gg r$$

$$\sqrt{L^2+r^2} \approx L$$

Hence,

$$A = \frac{\mu_0 I}{2\pi}\ln\left(\frac{2L}{r}\right)$$

$$\boxed{A = \frac{\mu_0 I}{2\pi}\ln\left(\frac{2L}{r}\right)}$$

Key Points

- Magnetic vector potential due to a current element is obtained from

$$d\vec{A} = \frac{\mu_0 I\, d\vec{l}}{4\pi R}.$$

- Trigonometric substitution

$$x = r\tan\theta$$

  simplifies the integral.
- For a long conductor,

$$L \gg r,$$

  the approximation

$$\sqrt{L^2+r^2} \approx L$$

  is used.

Q2.95.2 MCQ 1995 1M Ans: A

For a uniform plane electromagnetic wave propagating in the positive x-direction,

$$\vec{E} \times \vec{H} = \vec{S}$$

where $\vec{S}$ is the Poynting vector and indicates the direction of propagation.
Given,

$$\vec{E}(x,t) = E_0\cos(kx - \omega t)\,\hat{a}_y$$

For propagation along $+\hat{a}_x$,

$$\hat{a}_y \times \hat{a}_z = \hat{a}_x$$

Hence the magnetic field must be along the positive z-direction.
Also, in a lossless medium (vacuum), electric and magnetic fields are in phase.
Therefore,

$$\vec{H}(x,t) = H_0 \cos(kx - \omega t)\, \hat{a}_z$$

Checking the options:

- Option A: Correct direction and in phase.
- Option B: Electric and magnetic fields are 90° out of phase, which is not possible for a uniform plane wave in vacuum.
- Option C: Gives
$$\hat{a}_y \times (-\hat{a}_z) = -\hat{a}_x$$
which corresponds to propagation in the negative x-direction.

Hence,

$$\boxed{A}$$

Key Points

- Direction of propagation is given by the Poynting vector:
$$\vec{S} = \vec{E} \times \vec{H}.$$
- In vacuum, $\vec{E}$ and $\vec{H}$ are mutually perpendicular and are in phase.
- For propagation along $+\hat{a}_x$,
$$\hat{a}_y \times \hat{a}_z = \hat{a}_x.$$
- Magnitudes are related by
$$E = \eta_0 H,$$
where η_0 is the intrinsic impedance of free space.

Q2.94.1 NAT 1994 1M Ans: 2 ● ○ ○

The inductance of a coil is given by

$$L = \frac{\mu N^2 A}{l}$$

where

- N = number of turns,
- μ = permeability of the medium,
- A = cross-sectional area,
- l = magnetic path length.

If all other parameters remain constant, then

$$L \propto N^2$$

Hence, the inductance is proportional to the **square of the number of turns**.

$$\boxed{2}$$

Q2.94.2 MCQ | 1994 | 1M | Ans: a:R,b:S,c:P,d:Q

The matching is based on the most appropriate association of each item.

$$(a)\ \text{Line charge} \longrightarrow (R)\ \text{Transmission line conductors}$$

Transmission line conductors are commonly modeled as line charges for electric field and capacitance calculations.

$$(b)\ \text{Magnetic flux density} \longrightarrow (S)\ \text{Biot–Savart's law}$$

Biot–Savart's law is used to determine the magnetic flux density produced by current-carrying conductors.

$$(c)\ \text{Displacement current} \longrightarrow (P)\ \text{Maxwell}$$

Displacement current was introduced by Maxwell to modify Ampere's law and account for time-varying electric fields.

$$(d)\ \text{Power flow} \longrightarrow (Q)\ \text{Poynting's Vector}$$

Poynting's vector represents electromagnetic power flow density.
Hence, the correct matching is

$$\boxed{(a) - R,\ (b) - S,\ (c) - P,\ (d) - Q}$$

Key Points

- Transmission line conductors are modeled as line charges for electrostatic field analysis.
- Biot–Savart's law gives magnetic flux density:

$$d\vec{B} = \frac{\mu_0 I\, d\vec{l} \times \hat{r}}{4\pi r^2}$$

- Maxwell introduced the concept of displacement current:

$$J_d = \frac{\partial D}{\partial t}$$

- Poynting vector represents electromagnetic power flow:

$$\vec{S} = \vec{E} \times \vec{H}$$

Mistakes to be avoided

- Do not match line charge with Gauss's law in this association-type question; the expected match is the physical application, i.e., transmission line conductors.
- Do not confuse displacement current with conduction current.
- Remember that the Poynting vector represents power flow, not stored energy.
- Biot–Savart's law gives magnetic flux density, whereas Faraday's law deals with induced emf.

Q2.94.3 MCQ 1994 1M Ans: False ● ○ ○

A static magnetic field cannot induce current in a closed conducting loop because the magnetic flux linked with the loop remains constant.
According to Faraday's law of electromagnetic induction,

$$V_{\text{induced}} = -\frac{d\Phi}{dt}$$

For an induced emf (and hence current) to exist,

$$\frac{d\Phi}{dt} \neq 0$$

This means that the magnetic flux crossing the loop must vary with time.
For a static magnetic field,

$$\frac{d\Phi}{dt} = 0$$

Therefore,

$$V_{\text{induced}} = 0$$

Hence, no current is induced in the closed conducting loop.
Therefore, the given statement is

$$\boxed{\text{False}}$$

Q2.93.1 MCQ 1993 1M Ans: A ● ○ ○

The magnetic vector potential $\vec{A}$ is related to the magnetic flux density $\vec{B}$ by

$$\vec{B} = \nabla \times \vec{A}$$

Applying Stokes' theorem,

$$\oint_C \vec{A} \cdot d\vec{l} = \iint_S (\nabla \times \vec{A}) \cdot d\vec{S}$$

Substituting

$$\nabla \times \vec{A} = \vec{B}$$

gives

$$\oint_C \vec{A} \cdot d\vec{l} = \iint_S \vec{B} \cdot d\vec{S}$$

The right-hand side represents the total magnetic flux through the surface S.
Hence,

$$\oint_C \vec{A} \cdot d\vec{l} = \Phi$$

Therefore, the line integral of the vector potential around the boundary of a surface represents the magnetic flux through that surface.

$$\boxed{A}$$

Key Points

- Magnetic vector potential:

$$\vec{B} = \nabla \times \vec{A}$$

- Stokes' theorem:

$$\oint_C \vec{A} \cdot d\vec{l} = \iint_S (\nabla \times \vec{A}) \cdot d\vec{S}$$

- The surface integral

$$\iint_S \vec{B} \cdot d\vec{S}$$

  gives the magnetic flux.
- The circulation of $\vec{A}$ around a closed contour equals the magnetic flux enclosed by that contour.

DEBUG INFO

Chapter II

Magnetostatics

Total Questions : 41

MCQ : 24

MSQ : 1

NAT : 16